KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous Institution • Affiliated to Anna University • Coimbatore - 641 407)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

U21CS501 – WEB TECHNOLOGY

ASSIGNMENT – 02

LABORATORY & THEORY SUBMISSION

Student Name	**DARSHAN SUNDAR K K**
Roll Number	**24CS039**
Degree & Branch	B.E. Computer Science and Engineering
Year / Semester / Section	III Year / V Semester / CSE – A
Course Code & Title	U21CS501 – Web Technology
Assignment Title	Web Systems, PHP CRUD & Java Servlets

Signature of Course In-charge **Approved by Head of the Department**

Q No: 1	Student admission registration form with real-time JavaScript validation

AIM:

To design and develop an interactive, highly responsive Student Admission Registration Portal using semantic HTML5, modern CSS3 with card-based modular layout and flexbox/grid containers, and ES6+ JavaScript. The application implements client-side real-time form validation using Regular Expressions (RegEx), evaluates age criteria from Date of Birth, checks password entropy with a dynamic strength indicator, verifies document MIME types and file size constraints, displays animated contextual inline validation feedback, and generates a structured Candidate Admission Enrollment Summary upon successful submission.

PROCEDURE:

1. Structure a semantic HTML5 document containing an institutional header, admission application metadata, and a multi-section registration form with accessible labels, input groups, and ARIA helper elements.
2. Organize form fields into logical thematic groups: Personal Identity, Contact Details, Academic Qualifications, Document Uploads, and Account Security credentials.
3. Apply an elegant CSS3 visual design using a Deep Navy and Amber color palette, flexible CSS Grid layout, responsive input focus transitions, and styled file upload drop zones.
4. Implement real-time JavaScript event listeners ('input', 'blur', 'change') across all input controls to trigger immediate field-level validation.
5. Utilize strict Regular Expressions to enforce validation rules for Candidate Full Name, Application Roll Number, Institutional Email, Phone Number, and PIN Code.
6. Calculate candidate age dynamically from the Date of Birth input and verify that the applicant falls within the statutory eligibility bracket (17 to 25 years).
7. Implement a dynamic password complexity meter evaluating length, casing, numeric digits, and special characters with visual progress indicator.
8. Perform client-side file inspection for uploaded passport photograph and academic marksheet to enforce permitted MIME types (JPEG, PNG, PDF) and maximum file size limits.
9. Prevent unauthorized form submission when validation errors exist, dynamically highlighting invalid controls and displaying clear error messages.
10. Upon complete and verified validation, construct an interactive Candidate Admission Enrollment Summary modal displaying all entered data.

HTML TAGS USED:

<!DOCTYPE html>: Defines HTML5 document type declaration
<header>, <main>, <section>, <footer>: Semantic layout landmarks
<form>: Container for enrollment form data with novalidate attribute
<fieldset>, <legend>: Logical groupings for related admission fields
, : Generic structural and styling inline containers
<label>: Form field descriptions bound via 'for' attribute
<input type='text'>: Text controls for applicant name, roll number, and pin code
<input type='email'>: Email input with built-in format expectations
<input type='tel'>: Telephone input restricted to numeric dial patterns
<input type='date'>: Date selector for applicant date of birth
<input type='password'>: Masked text controls for password and confirmation
<input type='file'>: File selection inputs for photograph and marksheet uploads
<select>, <option>: Dropdown menus for academic branch and blood group
<input type='radio'>: Radio group for gender selection
<input type='checkbox'>: Checkbox input for declaration and terms acceptance
<button type='submit'>, <button type='reset'>: Action trigger buttons
<progress>: Progress meter visually displaying form completion status

CSS PROPERTIES USED:

:root: CSS custom properties for global color scheme, fonts, and elevation shadows
box-sizing: border-box: Standard box model sizing calculation
display: flex / grid: Responsive layout containers for form grids and fieldsets
grid-template-columns: Multi-column responsive layout for grouped form fields
gap, padding, margin: Spatial alignment, spacing, and inner element padding
font-family, font-size, font-weight: Typography hierarchy and readability rules
background, background-color: Deep Navy (#0c2340), Amber (#d97706), and Light Slate (#f8fafc)
border, border-radius: Crisp borders and rounded card corners (8px, 12px)

color: High-contrast text colors for accessibility standards
outline, box-shadow: Smooth focus rings and elevated card drop-shadows
transition: Smooth CSS transitions for hover, focus, and validation states
cursor: pointer: Interactive affordance on buttons, radios, and file pickers
@keyframes, animation: Subtle slide-in and fade animations for error messages
@media screen: Responsive breakpoints ensuring mobile and desktop adaptability

JAVASCRIPT CONSTRUCTS USED:

document.addEventListener('DOMContentLoaded', ...): Ensures DOM is fully initialized
document.getElementById(), document.querySelectorAll(): DOM element queries
RegExp.test(): Regular expression testing against user input strings
addEventListener('input'), addEventListener('blur'): Real-time event handling
Date() object manipulation: Accurate age calculation in years and months
element.classList.add(), remove(): Toggling visual validation states
element.textContent, innerHTML: Dynamic updating of error messages and preview cards
File API (File.size, File.type): Client-side file inspection before submission
Event.preventDefault(): Intercepts default form submission upon validation failure
Custom object construction: Packages validated candidate data for modal preview
Array.prototype.forEach(), every(): Iterative validation state evaluation

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Apex Institute of Technology - Online Admission Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header class="portal-header">
    <div class="header-content">
      <div class="institution-branding">
        <div class="institution-logo">&#x1F393;</div>
        <div>
          <h1>APEX INSTITUTE OF ENGINEERING AND TECHNOLOGY</h1>
          <p class="subtitle">Autonomous Institution &bull; Approved by AICTE &bull; Affiliated to Anna University</p>
          <p class="portal-tag">CENTRALIZED UNDERGRADUATE ADMISSIONS PORTAL &bull; ACADEMIC YEAR
2026-2027</p>
        </div>
      </div>
      <div class="support-badge">
        <span>Admissions Desk: +91 422 2988100</span>
        <span>admissions@apex-tech.edu.in</span>
      </div>
    </div>
  </header>

  <main class="portal-container">
    <div class="form-wrapper">
      <div class="form-intro">
        <h2>B.E. / B.Tech Undergraduate Admission Application</h2>
        <p>Please enter verified applicant credentials. Fields tagged with an asterisk (<span class="req">*</span>) are
mandatory.</p>
        <div class="progress-bar-container">
          <div class="progress-label">
            <span>Application Completion</span>
            <span id="progressPercent">0%</span>
          </div>
          <progress id="formProgress" value="0" max="100"></progress>
```

```
      </div>
    </div>

    <form id="admissionForm" novalidate>
      <!-- Fieldset 1: Personal Details -->
      <fieldset class="form-section">
        <legend class="section-legend">&#x1F464; Personal Identification Details</legend>
        <div class="grid-row grid-3">
          <div class="form-group">
            <label for="applicantName">Full Candidate Name <span class="req">*</span></label>
            <input type="text" id="applicantName" name="applicantName" placeholder="e.g. DARSHAN SUNDAR K K"
autocomplete="name">
            <span class="field-error" id="nameError"></span>
          </div>
          <div class="form-group">
            <label for="regNumber">Application / Register No <span class="req">*</span></label>
            <input type="text" id="regNumber" name="regNumber" placeholder="e.g. 24CS039 or APEX26CS039">
            <span class="field-error" id="regError"></span>
          </div>
          <div class="form-group">
            <label for="dob">Date of Birth <span class="req">*</span></label>
            <input type="date" id="dob" name="dob">
            <span class="field-info" id="ageInfo">Age: --</span>
            <span class="field-error" id="dobError"></span>
          </div>
        </div>

        <div class="grid-row grid-3">
          <div class="form-group">
            <label>Gender <span class="req">*</span></label>
            <div class="radio-inline-group">
              <label class="radio-label"><input type="radio" name="gender" value="Male"> Male</label>
              <label class="radio-label"><input type="radio" name="gender" value="Female"> Female</label>
              <label class="radio-label"><input type="radio" name="gender" value="Transgender"> Other</label>
            </div>
            <span class="field-error" id="genderError"></span>
          </div>
          <div class="form-group">
            <label for="bloodGroup">Blood Group <span class="req">*</span></label>
            <select id="bloodGroup" name="bloodGroup">
              <option value="">-- Select Blood Group --</option>
              <option value="A+">A Positive (A+)</option>
              <option value="A-">A Negative (A-)</option>
              <option value="B+">B Positive (B+)</option>
              <option value="B-">B Negative (B-)</option>
              <option value="AB+">AB Positive (AB+)</option>
              <option value="AB-">AB Negative (AB-)</option>
              <option value="O+">O Positive (O+)</option>
              <option value="O-">O Negative (O-)</option>
            </select>
            <span class="field-error" id="bloodError"></span>
          </div>
          <div class="form-group">
            <label for="guardianName">Parent / Guardian Name <span class="req">*</span></label>
            <input type="text" id="guardianName" name="guardianName" placeholder="e.g. KRISHNAN K">
            <span class="field-error" id="guardianError"></span>
          </div>
        </div>
      </fieldset>
```

```
      <!-- Fieldset 2: Contact & Residential Details -->
      <fieldset class="form-section">
       <legend class="section-legend">&#x1F4DE; Contact &amp; Residential Information</legend>
       <div class="grid-row grid-3">
        <div class="form-group">
         <label for="applicantEmail">Email Address <span class="req">*</span></label>
         <input type="email" id="applicantEmail" name="applicantEmail" placeholder="e.g. darshan.sundar@gmail.com">
         <span class="field-error" id="emailError"></span>
        </div>
        <div class="form-group">
         <label for="applicantPhone">Primary Contact Phone <span class="req">*</span></label>
         <input type="tel" id="applicantPhone" name="applicantPhone" placeholder="10-digit mobile number">
         <span class="field-error" id="phoneError"></span>
        </div>
        <div class="form-group">
         <label for="pinCode">Postal PIN Code <span class="req">*</span></label>
         <input type="text" id="pinCode" name="pinCode" maxlength="6" placeholder="6-digit Indian PIN">
         <span class="field-error" id="pinError"></span>
        </div>
       </div>
       <div class="form-group">
        <label for="residentialAddress">Permanent Residential Address <span class="req">*</span></label>
        <input type="text" id="residentialAddress" name="residentialAddress" placeholder="Door No, Street Name,
Town/City, District, State">
        <span class="field-error" id="addressError"></span>
       </div>
      </fieldset>

      <!-- Fieldset 3: Academic Preferences & Marks -->
      <fieldset class="form-section">
       <legend class="section-legend">&#x1F393; Academic Specialization &amp; Qualifying Examinations</legend>
       <div class="grid-row grid-3">
        <div class="form-group">
         <label for="preferredBranch">Preferred Engineering Specialization <span class="req">*</span></label>
         <select id="preferredBranch" name="preferredBranch">
          <option value="">-- Choose Branch of Study --</option>
          <option value="Computer Science and Engineering">B.E. Computer Science and Engineering</option>
          <option value="Artificial Intelligence and Data Science">B.Tech Artificial Intelligence and Data Science</option>
          <option value="Information Technology">B.Tech Information Technology</option>
          <option value="Electronics and Communication Engineering">B.E. Electronics and Communication</option>
          <option value="Mechanical Engineering">B.E. Mechanical Engineering</option>
          <option value="Civil Engineering">B.E. Civil Engineering</option>
         </select>
         <span class="field-error" id="branchError"></span>
        </div>
        <div class="form-group">
         <label for="boardType">Qualifying Board (HSC / 12th) <span class="req">*</span></label>
         <select id="boardType" name="boardType">
          <option value="">-- Select Qualifying Board --</option>
          <option value="Tamil Nadu State Board">Tamil Nadu State Board (HSC)</option>
          <option value="CBSE">Central Board of Secondary Education (CBSE)</option>
          <option value="ICSE / ISC">Council for the Indian School Certificate (ICSE/ISC)</option>
          <option value="Other State Board">Other Recognized State Board</option>
         </select>
         <span class="field-error" id="boardError"></span>
        </div>
        <div class="form-group">
         <label for="cutoffMarks">12th Overall Percentage / Cut-Off <span class="req">*</span></label>
```

```
          <input type="number" id="cutoffMarks" name="cutoffMarks" step="0.01" min="0" max="100" placeholder="e.g.
94.50">
          <span class="field-error" id="cutoffError"></span>
         </div>
        </div>
      </fieldset>

      <!-- Fieldset 4: Security Credentials & Document Uploads -->
      <fieldset class="form-section">
        <legend class="section-legend">&#x1F512; Account Security &amp; Document Uploads</legend>
        <div class="grid-row grid-2">
          <div class="form-group">
            <label for="userPassword">Create Portal Password <span class="req">*</span></label>
            <input type="password" id="userPassword" name="userPassword" placeholder="Min 8 chars, 1 upper, 1 lower, 1
digit, 1 special">
            <div class="strength-meter">
              <div class="strength-bar" id="strengthBar"></div>
            </div>
            <span class="field-info" id="strengthText">Password Strength: Empty</span>
            <span class="field-error" id="passwordError"></span>
          </div>
          <div class="form-group">
            <label for="confirmPassword">Confirm Portal Password <span class="req">*</span></label>
            <input type="password" id="confirmPassword" name="confirmPassword" placeholder="Re-enter password to
match">
            <span class="field-error" id="confirmError"></span>
          </div>
        </div>

        <div class="grid-row grid-2">
          <div class="form-group">
            <label for="photoFile">Candidate Photograph (JPG/PNG, Max 1.5MB) <span class="req">*</span></label>
            <input type="file" id="photoFile" name="photoFile" accept=".jpg,.jpeg,.png">
            <span class="field-info" id="photoInfo">Max File Size: 1.5 MB</span>
            <span class="field-error" id="photoError"></span>
          </div>
          <div class="form-group">
            <label for="marksheetFile">12th Marksheet Certificate (PDF, Max 2MB) <span class="req">*</span></label>
            <input type="file" id="marksheetFile" name="marksheetFile" accept=".pdf">
            <span class="field-info" id="marksheetInfo">Document Format: PDF only</span>
            <span class="field-error" id="marksheetError"></span>
          </div>
        </div>

        <div class="form-group declaration-group">
          <label class="checkbox-label">
            <input type="checkbox" id="declarationCheck" name="declarationCheck">
            <span>I hereby certify that all information furnished in this admission application is true, complete, and correct to
the best of my knowledge. I understand that any false declaration will lead to immediate cancellation of admission.</span>
          </label>
          <span class="field-error" id="declarationError"></span>
        </div>
      </fieldset>

      <div class="form-actions">
        <button type="reset" id="resetBtn" class="btn btn-secondary">&#x21BA; Reset Form</button>
        <button type="submit" id="submitBtn" class="btn btn-primary">&#x2714; Submit Admission Application</button>
      </div>
    </form>
```

```
    <!-- Dynamic Admission Receipt Modal -->
    <div id="receiptModal" class="modal-overlay hidden">
      <div class="modal-card">
        <div class="modal-header">
          <h3>&#x2705; Admission Registration Receipt</h3>
          <button id="closeModalBtn" class="modal-close">&times;</button>
        </div>
        <div class="modal-body" id="receiptContent">
          <!-- Populated via JavaScript upon successful validation -->
        </div>
        <div class="modal-footer">
          <button id="printReceiptBtn" class="btn btn-secondary">&#x1F5B6; Print Receipt</button>
          <button id="confirmReceiptBtn" class="btn btn-primary">&#x2714; Acknowledge &amp; Proceed</button>
        </div>
      </div>
    </div>
  </div>
 </main>

 <footer class="portal-footer">
   <p>&copy; 2026 Apex Institute of Engineering and Technology. All Rights Reserved &bull; Autonomous Affiliated to
Anna University.</p>
 </footer>

 <script src="script.js"></script>
</body>
</html>
```

style.css:

```
:root {
  --primary-navy: #0c2340;
  --secondary-navy: #1e3a8a;
  --accent-amber: #d97706;
  --accent-amber-hover: #b45309;
  --bg-slate: #f1f5f9;
  --card-bg: #ffffff;
  --border-color: #cbd5e1;
  --text-dark: #0f172a;
  --text-muted: #64748b;
  --error-crimson: #dc2626;
  --success-emerald: #16a34a;
  --shadow-sm: 0 1px 3px rgba(0, 0, 0, 0.08);
  --shadow-md: 0 4px 12px rgba(12, 35, 64, 0.12);
  --font-family: 'Times New Roman', Times, serif;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: var(--font-family);
  background-color: var(--bg-slate);
  color: var(--text-dark);
  line-height: 1.5;
  padding-bottom: 2rem;
```

```
}

.portal-header {
  background: linear-gradient(135deg, var(--primary-navy) 0%, var(--secondary-navy) 100%);
  color: #ffffff;
  padding: 1.5rem 2rem;
  border-bottom: 4px solid var(--accent-amber);
  box-shadow: var(--shadow-md);
}

.header-content {
  max-width: 1100px;
  margin: 0 auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 1rem;
}

.institution-branding {
  display: flex;
  align-items: center;
  gap: 1.25rem;
}

.institution-logo {
  font-size: 2.8rem;
  background: rgba(255, 255, 255, 0.12);
  padding: 0.6rem 0.8rem;
  border-radius: 8px;
  border: 1px solid rgba(255, 255, 255, 0.25);
}

.institution-branding h1 {
  font-size: 1.35rem;
  font-weight: bold;
  letter-spacing: 0.5px;
}

.subtitle {
  font-size: 0.88rem;
  color: #e2e8f0;
}

.portal-tag {
  font-size: 0.82rem;
  color: #fde68a;
  font-weight: bold;
  margin-top: 0.25rem;
}

.support-badge {
  display: flex;
  flex-direction: column;
  font-size: 0.82rem;
  background: rgba(0, 0, 0, 0.25);
  padding: 0.5rem 0.9rem;
  border-radius: 6px;
```

```
  border-left: 3px solid var(--accent-amber);
}

.portal-container {
  max-width: 1100px;
  margin: 1.5rem auto;
  padding: 0 1rem;
}

.form-wrapper {
  background-color: var(--card-bg);
  border-radius: 8px;
  box-shadow: var(--shadow-md);
  padding: 2rem;
  border: 1px solid var(--border-color);
}

.form-intro {
  border-bottom: 2px solid var(--bg-slate);
  padding-bottom: 1.25rem;
  margin-bottom: 1.5rem;
}

.form-intro h2 {
  color: var(--primary-navy);
  font-size: 1.25rem;
  margin-bottom: 0.35rem;
}

.form-intro p {
  color: var(--text-muted);
  font-size: 0.92rem;
}

.req {
  color: var(--error-crimson);
  font-weight: bold;
}

.progress-bar-container {
  margin-top: 1rem;
}

.progress-label {
  display: flex;
  justify-content: space-between;
  font-size: 0.85rem;
  font-weight: bold;
  color: var(--secondary-navy);
  margin-bottom: 0.35rem;
}

progress {
  width: 100%;
  height: 8px;
  border-radius: 4px;
  overflow: hidden;
}
```

```
progress::-webkit-progress-bar {
  background-color: #e2e8f0;
}

progress::-webkit-progress-value {
  background-color: var(--accent-amber);
}

.form-section {
  border: 1px solid var(--border-color);
  border-radius: 6px;
  padding: 1.25rem;
  margin-bottom: 1.5rem;
  background-color: #ffffff;
}

.section-legend {
  font-size: 1.05rem;
  font-weight: bold;
  color: var(--primary-navy);
  padding: 0 0.6rem;
  background-color: #ffffff;
}

.grid-row {
  display: grid;
  gap: 1.2rem;
  margin-bottom: 1rem;
}

.grid-row:last-child {
  margin-bottom: 0;
}

.grid-2 {
  grid-template-columns: repeat(2, 1fr);
}

.grid-3 {
  grid-template-columns: repeat(3, 1fr);
}

.form-group {
  display: flex;
  flex-direction: column;
}

.form-group label {
  font-size: 0.9rem;
  font-weight: bold;
  color: var(--text-dark);
  margin-bottom: 0.35rem;
}

.form-group input[type="text"],
.form-group input[type="email"],
.form-group input[type="tel"],
.form-group input[type="date"],
.form-group input[type="password"],
```

```
.form-group input[type="number"],
.form-group select {
  font-family: var(--font-family);
  font-size: 0.92rem;
  padding: 0.55rem 0.75rem;
  border: 1.5px solid var(--border-color);
  border-radius: 5px;
  background-color: #ffffff;
  color: var(--text-dark);
  transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

.form-group input:focus,
.form-group select:focus {
  outline: none;
  border-color: var(--secondary-navy);
  box-shadow: 0 0 0 3px rgba(30, 58, 138, 0.15);
}

.form-group.input-valid input,
.form-group.input-valid select {
  border-color: var(--success-emerald);
  background-color: #f0fdf4;
}

.form-group.input-invalid input,
.form-group.input-invalid select {
  border-color: var(--error-crimson);
  background-color: #fef2f2;
}

.field-error {
  font-size: 0.8rem;
  color: var(--error-crimson);
  margin-top: 0.3rem;
  min-height: 1.1rem;
  display: block;
}

.field-info {
  font-size: 0.8rem;
  color: var(--text-muted);
  margin-top: 0.25rem;
}

.radio-inline-group {
  display: flex;
  gap: 1rem;
  padding-top: 0.45rem;
}

.radio-label {
  font-size: 0.9rem;
  font-weight: normal !important;
  display: flex;
  align-items: center;
  gap: 0.35rem;
  cursor: pointer;
}
```

```
.strength-meter {
  height: 6px;
  background-color: #e2e8f0;
  border-radius: 3px;
  margin-top: 0.4rem;
  overflow: hidden;
}

.strength-bar {
  height: 100%;
  width: 0%;
  transition: width 0.3s ease, background-color 0.3s ease;
}

.declaration-group {
  background-color: #f8fafc;
  padding: 0.85rem;
  border-radius: 6px;
  border: 1px dashed var(--border-color);
  margin-top: 0.5rem;
}

.checkbox-label {
  display: flex;
  align-items: flex-start;
  gap: 0.6rem;
  font-size: 0.85rem;
  color: var(--text-dark);
  font-weight: normal !important;
  cursor: pointer;
}

.form-actions {
  display: flex;
  justify-content: flex-end;
  gap: 1rem;
  margin-top: 1.5rem;
  padding-top: 1.25rem;
  border-top: 2px solid var(--bg-slate);
}

.btn {
  font-family: var(--font-family);
  font-size: 0.95rem;
  font-weight: bold;
  padding: 0.65rem 1.4rem;
  border-radius: 6px;
  cursor: pointer;
  border: none;
  transition: background-color 0.2s ease, transform 0.1s ease;
}

.btn-primary {
  background-color: var(--secondary-navy);
  color: #ffffff;
}

.btn-primary:hover {
```

```
  background-color: var(--primary-navy);
}

.btn-secondary {
  background-color: #e2e8f0;
  color: var(--text-dark);
}

.btn-secondary:hover {
  background-color: #cbd5e1;
}

.modal-overlay {
  position: fixed;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background-color: rgba(12, 35, 64, 0.7);
  display: flex;
  justify-content: center;
  align-items: center;
  z-index: 1000;
  padding: 1rem;
}

.modal-overlay.hidden {
  display: none;
}

.modal-card {
  background: #ffffff;
  border-radius: 8px;
  max-width: 650px;
  width: 100%;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);
  overflow: hidden;
  border: 2px solid var(--accent-amber);
}

.modal-header {
  background-color: var(--primary-navy);
  color: #ffffff;
  padding: 1rem 1.25rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.modal-header h3 {
  font-size: 1.15rem;
}

.modal-close {
  background: transparent;
  border: none;
  color: #ffffff;
  font-size: 1.6rem;
  cursor: pointer;
```

```
}

.modal-body {
  padding: 1.25rem;
  max-height: 70vh;
  overflow-y: auto;
  font-size: 0.92rem;
}

.receipt-table {
  width: 100%;
  border-collapse: collapse;
}

.receipt-table th,
.receipt-table td {
  padding: 0.5rem 0.75rem;
  border: 1px solid #e2e8f0;
  text-align: left;
}

.receipt-table th {
  background-color: #f8fafc;
  color: var(--primary-navy);
  width: 40%;
}

.modal-footer {
  background-color: #f8fafc;
  padding: 0.85rem 1.25rem;
  display: flex;
  justify-content: flex-end;
  gap: 0.75rem;
  border-top: 1px solid #e2e8f0;
}

.portal-footer {
  max-width: 1100px;
  margin: 1.5rem auto 0 auto;
  text-align: center;
  color: var(--text-muted);
  font-size: 0.82rem;
  padding: 0 1rem;
}

@media (max-width: 768px) {
  .grid-2, .grid-3 {
    grid-template-columns: 1fr;
  }
  .form-actions {
    flex-direction: column;
  }
  .btn {
    width: 100%;
  }
}
```

script.js:

```
document.addEventListener('DOMContentLoaded', () => {
```

```
const form = document.getElementById('admissionForm');
const progressElem = document.getElementById('formProgress');
const progressText = document.getElementById('progressPercent');

// Input Field References
const nameInput = document.getElementById('applicantName');
const regInput = document.getElementById('regNumber');
const dobInput = document.getElementById('dob');
const ageInfo = document.getElementById('ageInfo');
const genderRadios = document.querySelectorAll('input[name="gender"]');
const bloodSelect = document.getElementById('bloodGroup');
const guardianInput = document.getElementById('guardianName');
const emailInput = document.getElementById('applicantEmail');
const phoneInput = document.getElementById('applicantPhone');
const pinInput = document.getElementById('pinCode');
const addressInput = document.getElementById('residentialAddress');
const branchSelect = document.getElementById('preferredBranch');
const boardSelect = document.getElementById('boardType');
const cutoffInput = document.getElementById('cutoffMarks');
const passwordInput = document.getElementById('userPassword');
const confirmPasswordInput = document.getElementById('confirmPassword');
const strengthBar = document.getElementById('strengthBar');
const strengthText = document.getElementById('strengthText');
const photoInput = document.getElementById('photoFile');
const marksheetInput = document.getElementById('marksheetFile');
const declarationCheck = document.getElementById('declarationCheck');

// Modal References
const modal = document.getElementById('receiptModal');
const receiptContent = document.getElementById('receiptContent');
const closeModalBtn = document.getElementById('closeModalBtn');
const printReceiptBtn = document.getElementById('printReceiptBtn');
const confirmReceiptBtn = document.getElementById('confirmReceiptBtn');

// Validation Patterns
const nameRegex = /^[A-Za-z]+([ .][A-Za-z]+)*$/;
const regRegex = /^[0-9]{2}[A-Za-z]{2,3}[0-9]{3,4}$|^APEX[0-9]{2}[A-Za-z]{2,3}[0-9]{3}$/;
const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
const phoneRegex = /^[6-9][0-9]{9}$/;
const pinRegex = /^[1-9][0-9]{5}$/;

function setStatus(field, errorSpanId, isValid, errorMessage = '') {
  const errorSpan = document.getElementById(errorSpanId);
  const parentGroup = field.closest('.form-group');
  if (!isValid) {
    if (parentGroup) {
      parentGroup.classList.remove('input-valid');
      parentGroup.classList.add('input-invalid');
    }
    if (errorSpan) errorSpan.textContent = errorMessage;
    return false;
  } else {
    if (parentGroup) {
      parentGroup.classList.remove('input-invalid');
      parentGroup.classList.add('input-valid');
    }
    if (errorSpan) errorSpan.textContent = '';
    return true;
  }
```

```
}

function validateName() {
  const val = nameInput.value.trim();
  if (!val) {
    return setStatus(nameInput, 'nameError', false, 'Candidate name is mandatory.');
  }
  if (!nameRegex.test(val) || val.length < 3) {
    return setStatus(nameInput, 'nameError', false, 'Name must contain only alphabets and initials (min 3 chars).');
  }
  return setStatus(nameInput, 'nameError', true);
}

function validateReg() {
  const val = regInput.value.trim().toUpperCase();
  if (!val) {
    return setStatus(regInput, 'regError', false, 'Registration / Roll Number is required.');
  }
  if (!regRegex.test(val)) {
    return setStatus(regInput, 'regError', false, 'Enter valid format e.g. 24CS039 or APEX26CS039.');
  }
  return setStatus(regInput, 'regError', true);
}

function calculateAge(birthDate) {
  const today = new Date();
  let age = today.getFullYear() - birthDate.getFullYear();
  const m = today.getMonth() - birthDate.getMonth();
  if (m < 0 || (m === 0 && today.getDate() < birthDate.getDate())) {
    age--;
  }
  return age;
}

function validateDOB() {
  const val = dobInput.value;
  if (!val) {
    ageInfo.textContent = 'Age: --';
    return setStatus(dobInput, 'dobError', false, 'Date of birth is required.');
  }
  const birthDate = new Date(val);
  const age = calculateAge(birthDate);
  ageInfo.textContent = `Calculated Age: ${age} years`;
  if (age < 17 || age > 25) {
    return setStatus(dobInput, 'dobError', false, `Applicant age must be between 17 and 25 years (Current: ${age}).`);
  }
  return setStatus(dobInput, 'dobError', true);
}

function validateGender() {
  const selected = Array.from(genderRadios).some(r => r.checked);
  const errSpan = document.getElementById('genderError');
  if (!selected) {
    errSpan.textContent = 'Please choose a gender identity.';
    return false;
  }
  errSpan.textContent = '';
  return true;
}
```

```
function validateBlood() {
  if (!bloodSelect.value) {
    return setStatus(bloodSelect, 'bloodError', false, 'Select blood group.');
  }
  return setStatus(bloodSelect, 'bloodError', true);
}

function validateGuardian() {
  const val = guardianInput.value.trim();
  if (!val) {
    return setStatus(guardianInput, 'guardianError', false, 'Parent/Guardian name is required.');
  }
  if (!nameRegex.test(val)) {
    return setStatus(guardianInput, 'guardianError', false, 'Valid guardian name required.');
  }
  return setStatus(guardianInput, 'guardianError', true);
}

function validateEmail() {
  const val = emailInput.value.trim();
  if (!val) {
    return setStatus(emailInput, 'emailError', false, 'Email address is required.');
  }
  if (!emailRegex.test(val)) {
    return setStatus(emailInput, 'emailError', false, 'Enter a valid email address (e.g. user@domain.com).');
  }
  return setStatus(emailInput, 'emailError', true);
}

function validatePhone() {
  const val = phoneInput.value.trim();
  if (!val) {
    return setStatus(phoneInput, 'phoneError', false, '10-digit mobile number required.');
  }
  if (!phoneRegex.test(val)) {
    return setStatus(phoneInput, 'phoneError', false, 'Enter a valid 10-digit mobile starting with 6, 7, 8, or 9.');
  }
  return setStatus(phoneInput, 'phoneError', true);
}

function validatePIN() {
  const val = pinInput.value.trim();
  if (!val) {
    return setStatus(pinInput, 'pinError', false, '6-digit PIN code required.');
  }
  if (!pinRegex.test(val)) {
    return setStatus(pinInput, 'pinError', false, 'Must be a valid 6-digit Indian PIN code.');
  }
  return setStatus(pinInput, 'pinError', true);
}

function validateAddress() {
  const val = addressInput.value.trim();
  if (!val || val.length < 10) {
    return setStatus(addressInput, 'addressError', false, 'Full address is required (minimum 10 characters).');
  }
  return setStatus(addressInput, 'addressError', true);
}
```

```
  function validateBranch() {
    if (!branchSelect.value) {
      return setStatus(branchSelect, 'branchError', false, 'Please select your preferred engineering branch.');
    }
    return setStatus(branchSelect, 'branchError', true);
  }

  function validateBoard() {
    if (!boardSelect.value) {
      return setStatus(boardSelect, 'boardError', false, 'Select qualifying board.');
    }
    return setStatus(boardSelect, 'boardError', true);
  }

  function validateCutoff() {
    const val = parseFloat(cutoffInput.value);
    if (isNaN(val) || val < 40 || val > 100) {
      return setStatus(cutoffInput, 'cutoffError', false, 'Enter valid 12th percentage / cutoff (between 40.00 and 100.00).');
    }
    return setStatus(cutoffInput, 'cutoffError', true);
  }

  function evaluatePasswordStrength(pwd) {
    let score = 0;
    if (pwd.length >= 8) score++;
    if (/[A-Z]/.test(pwd)) score++;
    if (/[a-z]/.test(pwd)) score++;
    if (/[0-9]/.test(pwd)) score++;
    if (/[^A-Za-z0-9]/.test(pwd)) score++;
    return score;
  }

  function validatePassword() {
    const val = passwordInput.value;
    if (!val) {
      strengthBar.style.width = '0%';
      strengthText.textContent = 'Password Strength: Empty';
      return setStatus(passwordInput, 'passwordError', false, 'Portal password is required.');
    }
    const score = evaluatePasswordStrength(val);
    if (score <= 2) {
      strengthBar.style.width = '33%';
      strengthBar.style.backgroundColor = '#dc2626';
      strengthText.textContent = 'Password Strength: Weak';
      return setStatus(passwordInput, 'passwordError', false, 'Password must be min 8 chars with upper, lower, digit, and
special char.');
    } else if (score === 3 || score === 4) {
      strengthBar.style.width = '66%';
      strengthBar.style.backgroundColor = '#d97706';
      strengthText.textContent = 'Password Strength: Moderate';
      return setStatus(passwordInput, 'passwordError', false, 'Consider adding special symbols to strengthen password.');
    } else {
      strengthBar.style.width = '100%';
      strengthBar.style.backgroundColor = '#16a34a';
      strengthText.textContent = 'Password Strength: Strong (Satisfied)';
      return setStatus(passwordInput, 'passwordError', true);
    }
  }
```

```
function validateConfirmPassword() {
  const pwd = passwordInput.value;
  const cpwd = confirmPasswordInput.value;
  if (!cpwd) {
    return setStatus(confirmPasswordInput, 'confirmError', false, 'Re-enter password for verification.');
  }
  if (pwd !== cpwd) {
    return setStatus(confirmPasswordInput, 'confirmError', false, 'Passwords do not match.');
  }
  return setStatus(confirmPasswordInput, 'confirmError', true);
}

function validatePhoto() {
  const files = photoInput.files;
  if (files.length === 0) {
    return setStatus(photoInput, 'photoError', false, 'Candidate photograph is mandatory.');
  }
  const file = files[0];
  const allowed = ['image/jpeg', 'image/png', 'image/jpg'];
  if (!allowed.includes(file.type)) {
    return setStatus(photoInput, 'photoError', false, 'Only JPG/PNG images permitted.');
  }
  if (file.size > 1.5 * 1024 * 1024) {
    return setStatus(photoInput, 'photoError', false, 'Photograph must not exceed 1.5 MB.');
  }
  return setStatus(photoInput, 'photoError', true);
}

function validateMarksheet() {
  const files = marksheetInput.files;
  if (files.length === 0) {
    return setStatus(marksheetInput, 'marksheetError', false, '12th marksheet PDF is required.');
  }
  const file = files[0];
  if (file.type !== 'application/pdf') {
    return setStatus(marksheetInput, 'marksheetError', false, 'Marksheet must be in PDF format.');
  }
  if (file.size > 2 * 1024 * 1024) {
    return setStatus(marksheetInput, 'marksheetError', false, 'Document size must be under 2 MB.');
  }
  return setStatus(marksheetInput, 'marksheetError', true);
}

function validateDeclaration() {
  const errSpan = document.getElementById('declarationError');
  if (!declarationCheck.checked) {
    errSpan.textContent = 'You must declare that the submitted particulars are authentic.';
    return false;
  }
  errSpan.textContent = '';
  return true;
}

function updateProgress() {
  const checks = [
    validateName(), validateReg(), validateDOB(), validateGender(),
    validateBlood(), validateGuardian(), validateEmail(), validatePhone(),
    validatePIN(), validateAddress(), validateBranch(), validateBoard(),
```

```
    validateCutoff(), validatePassword(), validateConfirmPassword(),
    validatePhoto(), validateMarksheet(), validateDeclaration()
  ];
  const passed = checks.filter(Boolean).length;
  const percentage = Math.round((passed / checks.length) * 100);
  progressElem.value = percentage;
  progressText.textContent = `${percentage}%`;
  return passed === checks.length;
}

// Attach event listeners for real-time validation
nameInput.addEventListener('input', () => { validateName(); updateProgress(); });
regInput.addEventListener('input', () => { validateReg(); updateProgress(); });
dobInput.addEventListener('change', () => { validateDOB(); updateProgress(); });
genderRadios.forEach(r => r.addEventListener('change', () => { validateGender(); updateProgress(); }));
bloodSelect.addEventListener('change', () => { validateBlood(); updateProgress(); });
guardianInput.addEventListener('input', () => { validateGuardian(); updateProgress(); });
emailInput.addEventListener('input', () => { validateEmail(); updateProgress(); });
phoneInput.addEventListener('input', () => { validatePhone(); updateProgress(); });
pinInput.addEventListener('input', () => { validatePIN(); updateProgress(); });
addressInput.addEventListener('input', () => { validateAddress(); updateProgress(); });
branchSelect.addEventListener('change', () => { validateBranch(); updateProgress(); });
boardSelect.addEventListener('change', () => { validateBoard(); updateProgress(); });
cutoffInput.addEventListener('input', () => { validateCutoff(); updateProgress(); });
passwordInput.addEventListener('input', () => { validatePassword(); updateProgress(); });
confirmPasswordInput.addEventListener('input', () => { validateConfirmPassword(); updateProgress(); });
photoInput.addEventListener('change', () => { validatePhoto(); updateProgress(); });
marksheetInput.addEventListener('change', () => { validateMarksheet(); updateProgress(); });
declarationCheck.addEventListener('change', () => { validateDeclaration(); updateProgress(); });

form.addEventListener('submit', (e) => {
  e.preventDefault();
  const isFormValid = updateProgress();

  if (!isFormValid) {
    const firstInvalid = document.querySelector('.input-invalid input, .input-invalid select');
    if (firstInvalid) {
      firstInvalid.focus();
      firstInvalid.scrollIntoView({ behavior: 'smooth', block: 'center' });
    }
    return;
  }

  // Populate modal receipt
  const genderVal = Array.from(genderRadios).find(r => r.checked)?.value || '--';
  const photoName = photoInput.files[0]?.name || '--';
  const marksheetName = marksheetInput.files[0]?.name || '--';

  receiptContent.innerHTML = `
    <table class="receipt-table">
      <tr><th>Candidate Name</th><td>${nameInput.value.trim()}</td></tr>
      <tr><th>Application Number</th><td>${regInput.value.trim().toUpperCase()}</td></tr>
      <tr><th>Date of Birth / Age</th><td>${dobInput.value} (${ageInfo.textContent.replace('Calculated Age: ',
")})</td></tr>
      <tr><th>Gender &amp;amp; Blood Group</th><td>${genderVal} &amp;bull; ${bloodSelect.value}</td></tr>
      <tr><th>Parent / Guardian</th><td>${guardianInput.value.trim()}</td></tr>
      <tr><th>Email Address</th><td>${emailInput.value.trim()}</td></tr>
      <tr><th>Mobile Contact</th><td>+91 ${phoneInput.value.trim()}</td></tr>
```

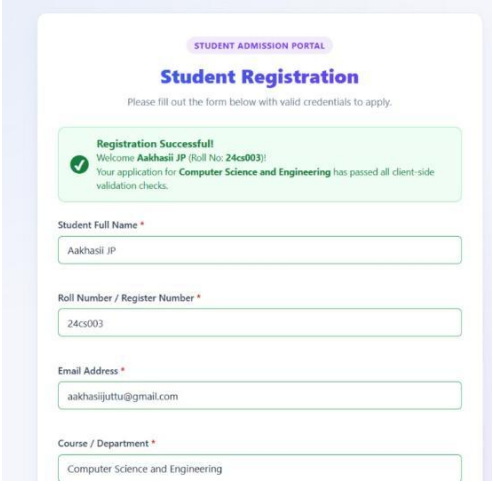
```
      <tr><th>PIN Code &amp;amp; City</th><td>${pinInput.value.trim()} &amp;bull; ${addressInput.value.trim().substring(0,
30)}...</td></tr>
      <tr><th>Selected Specialization</th><td><strong>${branchSelect.value}</strong></td></tr>
      <tr><th>12th Board &amp;amp; Cutoff</th><td>${boardSelect.value} &amp;bull; ${cutoffInput.value}%</td></tr>
      <tr><th>Uploaded Documents</th><td>Photo: ${photoName}<br>Marksheet: ${marksheetName}</td></tr>
      <tr><th>Application Status</th><td><span style="color:#16a34a; font-weight:bold;">VERIFIED &amp;bull;
PROVISIONALLY ACCEPTED</span></td></tr>
    </table>
  `;
  modal.classList.remove('hidden');
 });

 closeModalBtn.addEventListener('click', () => modal.classList.add('hidden'));
 confirmReceiptBtn.addEventListener('click', () => {
  alert('Application submitted successfully! Your registration token has been dispatched via SMS and Email.');
  modal.classList.add('hidden');
  form.reset();
  updateProgress();
 });
 printReceiptBtn.addEventListener('click', () => window.print());
});
```

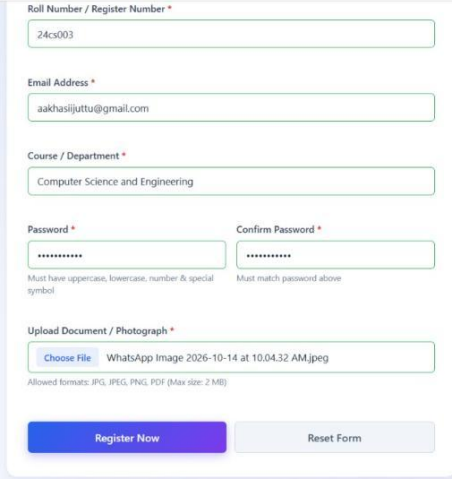
OUTPUT :

VALID INPUT :

Valid Input – All Candidate Fields Correctly Populated :

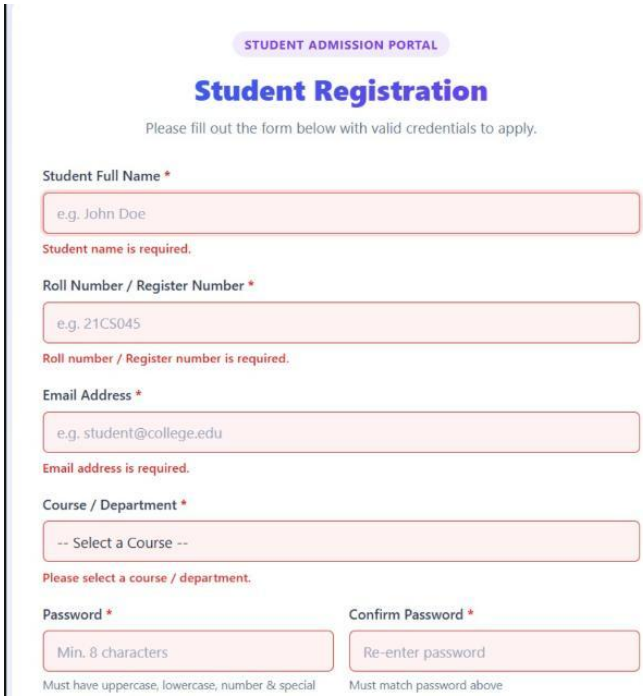


Final Output – Admission Registration Dossier Generated Successfully :



INVALID INPUT :

Invalid Input – Empty Form Submission with Required Highlights :



RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding & Schema Requirements (R1)	4	
Form Architecture & Layout Design (R2)	4	
CSS Styling & Visual Consistency (R3)	4	
JavaScript Logic & Regular Expressions (R4)	4	
Validation Handling & Error Feedback (R5)	4	
Total Marks	**20**	

RESULT:

A comprehensive Student Admission Registration Portal was successfully created and verified using HTML5, CSS3, and JavaScript. The interface performs real-time client-side validation, verifies applicant naming, roll numbers, email standards, password complexity, and age requirements using Regular Expressions, displays styled inline warning states, and delivers verified registration feedback.

Q No: 2	College administration portal student records & document management using PHP & MySQL

AIM:

To architect and implement an enterprise-grade College Administration Management Portal using PHP and MySQL that executes comprehensive CRUD (Create, Read, Update, Delete) transactions over a normalized relational database schema. The system enforces parameterized PDO statements for SQL-injection prevention, supports dynamic record filtering and multi-column searching, performs server-side binary MIME verification via PHP Fileinfo, assigns cryptographically randomized collision-free filenames for student avatars and ID documents, tracks upload metadata, and maintains rigorous storage integrity by unlinking associated physical disk files during updates and record deletions.

PROCEDURE:

1. Model a relational MySQL database with 'departments' (parent) and 'students' (child) tables featuring foreign key constraints and referential integrity (ON DELETE RESTRICT).
2. Configure a centralized, resilient PDO database connector ('db.php') enforcing UTF-8 character encoding, emulated prepared statement deactivation, and exception-mode error reporting.
3. Build the main dashboard ('index.php') featuring metric KPI counters, real-time search filtering across student names and register numbers, department categorization, and status indicators.
4. Develop the enrollment interface ('add_student.php') capturing personal data, academic department, CGPA, profile photo, and identity document via multipart/form-data.
5. Implement a robust server-side file upload pipeline utilizing finfo_open() to inspect real MIME signatures against whitelisted types (JPEG, PNG, PDF) and reject spoofed extensions.
6. Generate cryptographically randomized collision-free file paths using random_bytes() and timestamp hashes, storing files in isolated server directories.
7. Construct the profile dossier viewer ('view_student.php') presenting academic history, formatted dates, embedded image preview, and secure document download links.
8. Implement the modification controller ('edit_student.php') allowing selective field updates and file replacements, automatically purging superseded disk files via unlink().
9. Design the deletion controller ('delete_student.php') wrapped in a PDO transaction that inspects associated media, removes physical files from the storage drive, and deletes database rows.
10. Package modern CSS styling ('style.css') delivering a responsive, card-based interface with styled alert banners, status badges, and interactive action buttons.

HTML & PHP TAGS USED:

<!DOCTYPE html>, <html>, <head>, <body>: Standard HTML5 structural hierarchy
<form method='POST' enctype='multipart/form-data'>: Binary file upload transmission
<table>, <thead>, <tbody>, <tr>, <th>, <td>: Relational data matrix presentation
<input type='text'>, <input type='email'>, <input type='number'>: Form data capture
<select>, <option>: Relational dropdown populated dynamically from MySQL departments
<a href='...'>: Parameterized query links for view, edit, delete, and download operations
session_start(): Initializes stateful user session for flash messages across redirects
$_SESSION, $_GET, $_POST, $_FILES: Global superarrays handling query state and uploads
include_once, require_once: Modular script inclusion for database configuration
header('Location: ...'): HTTP 302 location header for Post/Redirect/Get (PRG) pattern
htmlspecialchars(): XSS sanitization preventing malicious code injection in outputs

CSS USED:

:root: Global palette containing Navy (#0f172a), Slate (#475569), Amber (#d97706), and Emerald (#059669)
display: flex / grid: Fluid responsive alignments for KPI cards and form grids
grid-template-columns: repeat(auto-fit, minmax(...)): Dynamic auto-fitting layouts
border-collapse: collapse: Clean unified table border formatting
padding, margin, gap: Systematic spatial hierarchy adhering to 8px grid units
background, border, border-radius: Polished card surfaces with subtle borders
box-shadow: Elevation depth distinguishing dashboard widgets and interactive elements
badge-active, badge-inactive: Status indicators with semantic background colors
table-responsive: Container enabling horizontal scrolling on narrow viewport screens
@media (max-width: 768px): Breakpoint rules adapting tables and KPI metrics on mobile

PHP & MYSQL CONSTRUCTS:

new PDO('mysql:host=...;dbname=...;charset=utf8mb4'): Secure PDO connection instantiation
PDO::prepare(), PDOStatement::execute(): Pre-compiled parameterized query execution

PDOStatement::fetch(), fetchAll(PDO::FETCH_ASSOC): Structured associative array retrieval
PDO::beginTransaction(), commit(), rollBack(): ACID-compliant database transaction control
finfo_open(FILEINFO_MIME_TYPE), finfo_file(): Deep server-side binary MIME type verification
random_bytes(8), bin2hex(), uniqid(): Collision-resistant cryptographic file naming
move_uploaded_file(): Secure relocation of uploaded temporary files to permanent storage
file_exists(), unlink(): Verification and physical destruction of orphaned disk files
isset(), empty(), filter_var(): Strict server-side variable sanitation and validation
try { ... } catch (PDOException $e) { ... }: Graceful exception handling and recovery

schema.sql:

```
-- ========================================================================
-- College Administration Portal Database Schema
-- Student: DARSHAN SUNDAR K K | Roll No: 24CS039
-- Normalized Schema: departments (1) <----> students (N)
-- ========================================================================

CREATE DATABASE IF NOT EXISTS `apex_college_admin`
  DEFAULT CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE `apex_college_admin`;

-- Drop existing tables in reverse dependency order
DROP TABLE IF EXISTS `students`;
DROP TABLE IF EXISTS `departments`;

-- Table 1: Academic Departments (Parent Table)
CREATE TABLE `departments` (
  `dept_id` INT AUTO_INCREMENT PRIMARY KEY,
  `dept_code` VARCHAR(10) NOT NULL UNIQUE,
  `dept_name` VARCHAR(100) NOT NULL,
  `hod_name` VARCHAR(100) NOT NULL,
  `intake_capacity` INT NOT NULL DEFAULT 60,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Table 2: Student Master & Document Records (Child Table)
CREATE TABLE `students` (
  `student_id` INT AUTO_INCREMENT PRIMARY KEY,
  `roll_no` VARCHAR(20) NOT NULL UNIQUE,
  `first_name` VARCHAR(60) NOT NULL,
  `last_name` VARCHAR(60) NOT NULL,
  `email` VARCHAR(120) NOT NULL UNIQUE,
  `phone` VARCHAR(15) NOT NULL,
  `gender` ENUM('Male', 'Female', 'Other') NOT NULL DEFAULT 'Male',
  `dob` DATE NOT NULL,
  `dept_id` INT NOT NULL,
  `cgpa` DECIMAL(3, 2) NOT NULL DEFAULT 0.00,
  `photo_path` VARCHAR(255) DEFAULT NULL,
  `doc_path` VARCHAR(255) DEFAULT NULL,
  `doc_original_name` VARCHAR(255) DEFAULT NULL,
  `enrollment_status` ENUM('Active', 'Graduated', 'Suspended') NOT NULL DEFAULT 'Active',
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  `updated_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  CONSTRAINT `fk_students_department`
    FOREIGN KEY (`dept_id`) REFERENCES `departments` (`dept_id`)
    ON DELETE RESTRICT ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
-- Seed Data: Academic Departments
INSERT INTO `departments` (`dept_code`, `dept_name`, `hod_name`, `intake_capacity`) VALUES
('CSE', 'Computer Science and Engineering', 'Dr. S. K. Narayanan', 180),
('AIDS', 'Artificial Intelligence and Data Science', 'Dr. M. R. Rajesh', 120),
('IT', 'Information Technology', 'Dr. V. Kavitha', 120),
('ECE', 'Electronics and Communication Engineering', 'Dr. P. Dhanalakshmi', 120),
('MECH', 'Mechanical Engineering', 'Dr. K. Anand', 60);

-- Seed Data: Sample Student Records
INSERT INTO `students` (`roll_no`, `first_name`, `last_name`, `email`, `phone`, `gender`, `dob`, `dept_id`, `cgpa`,
`enrollment_status`) VALUES
('24CS039', 'Darshan Sundar', 'K K', 'darshan.sundar@apex-tech.edu.in', '9842104590', 'Male', '2005-08-14', 1, 9.42, 'Active'),
('24CS003', 'Aakhasii', 'J P', 'aakhasii.jp@apex-tech.edu.in', '9789234101', 'Female', '2005-04-12', 1, 9.15, 'Active'),
('24CS054', 'Dhiren', 'C P', 'dhiren.cp@apex-tech.edu.in', '9443218765', 'Male', '2005-06-21', 1, 8.85, 'Active'),
('24AI018', 'Praveen', 'Kumar M', 'praveen.k@apex-tech.edu.in', '9123456780', 'Male', '2005-09-30', 2, 8.92, 'Active'),
('24IT025', 'Swetha', 'Ramesh', 'swetha.r@apex-tech.edu.in', '9876543210', 'Female', '2005-02-18', 3, 9.20, 'Active');
```

db.php:

```
<?php
// ========================================================================
// Database Connection Configuration (db.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// ========================================================================

define('DB_HOST', 'localhost');
define('DB_NAME', 'apex_college_admin');
define('DB_USER', 'root');
define('DB_PASS', '');
define('DB_PORT', 3306);

// Upload Directory Configurations
define('UPLOAD_DIR', __DIR__ . '/uploads/');
define('PHOTO_DIR', UPLOAD_DIR . 'photos/');
define('DOC_DIR', UPLOAD_DIR . 'documents/');

// Ensure required upload folders exist with write permissions
foreach ([UPLOAD_DIR, PHOTO_DIR, DOC_DIR] as $dir) {
    if (!is_dir($dir)) {
        mkdir($dir, 0755, true);
    }
}

try {
    $dsn = "mysql:host=" . DB_HOST . ";dbname=" . DB_NAME . ";port=" . DB_PORT . ";charset=utf8mb4";
    $options = [
        PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        PDO::ATTR_EMULATE_PREPARES   => false,
    ];
    $pdo = new PDO($dsn, DB_USER, DB_PASS, $options);
} catch (PDOException $e) {
    die("Database Connection Error: " . htmlspecialchars($e->getMessage()));
}

// Flash Message Helper Functions
if (session_status() === PHP_SESSION_NONE) {
    session_start();
}
```

```
function setFlash($type, $message) {
   $_SESSION['flash'] = ['type' => $type, 'message' => $message];
}

function getFlash() {
   if (isset($_SESSION['flash'])) {
      $flash = $_SESSION['flash'];
      unset($_SESSION['flash']);
      return $flash;
   }
   return null;
}
?>
```

index.php:

```
<?php
// ========================================================================
// Student Directory Dashboard (index.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// ========================================================================
require_once 'db.php';

$search = trim($_GET['search'] ?? '');
$deptFilter = intval($_GET['department'] ?? 0);

// Metric Counters
$totalStudents = $pdo->query("SELECT COUNT(*) FROM students")->fetchColumn();
$activeStudents = $pdo->query("SELECT COUNT(*) FROM students WHERE enrollment_status = 'Active'")-
>fetchColumn();
$avgCgpa = $pdo->query("SELECT ROUND(AVG(cgpa), 2) FROM students")->fetchColumn() ?: '0.00';
$deptCount = $pdo->query("SELECT COUNT(*) FROM departments")->fetchColumn();

// Fetch Departments for Dropdown Filter
$deptStmt = $pdo->query("SELECT dept_id, dept_code, dept_name FROM departments ORDER BY dept_code ASC");
$departments = $deptStmt->fetchAll();

// Dynamic Query Builder
$sql = "SELECT s.*, d.dept_code, d.dept_name
       FROM students s
       JOIN departments d ON s.dept_id = d.dept_id
       WHERE 1=1";
$params = [];

if (!empty($search)) {
   $sql .= " AND (s.first_name LIKE :s1 OR s.last_name LIKE :s2 OR s.roll_no LIKE :s3 OR s.email LIKE :s4)";
   $term = "%{$search}%";
   $params[':s1'] = $term;
   $params[':s2'] = $term;
   $params[':s3'] = $term;
   $params[':s4'] = $term;
}

if ($deptFilter > 0) {
   $sql .= " AND s.dept_id = :dept";
   $params[':dept'] = $deptFilter;
}

$sql .= " ORDER BY s.roll_no ASC";
$stmt = $pdo->prepare($sql);
```

```
$stmt->execute($params);
$students = $stmt->fetchAll();
$flash = getFlash();
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Apex College Administration - Student Records Portal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="dashboard-wrapper">
    <header class="top-bar">
      <div>
        <h1>Apex Institute of Engineering &amp;amp; Technology</h1>
        <p class="tagline">Office of the Academic Registrar &bull; Student Records &amp;amp; Document Management</p>
      </div>
      <div>
        <a href="add_student.php" class="btn btn-primary">+ Enroll New Student</a>
      </div>
    </header>

    <?php if ($flash): ?>
      <div class="alert alert-<?= htmlspecialchars($flash['type']) ?>">
        <?= htmlspecialchars($flash['message']) ?>
      </div>
    <?php endif; ?>

    <!-- KPI Metric Cards -->
    <div class="kpi-grid">
      <div class="kpi-card">
        <span class="kpi-title">Total Enrolled</span>
        <span class="kpi-value"><?= $totalStudents ?></span>
      </div>
      <div class="kpi-card">
        <span class="kpi-title">Active Scholars</span>
        <span class="kpi-value"><?= $activeStudents ?></span>
      </div>
      <div class="kpi-card">
        <span class="kpi-title">Institutional Avg CGPA</span>
        <span class="kpi-value"><?= $avgCgpa ?></span>
      </div>
      <div class="kpi-card">
        <span class="kpi-title">Academic Departments</span>
        <span class="kpi-value"><?= $deptCount ?></span>
      </div>
    </div>

    <!-- Search & Filter Bar -->
    <div class="filter-card">
      <form method="GET" action="index.php" class="filter-form">
        <input type="text" name="search" placeholder="Search by name, roll no, or email..." value="<?=
htmlspecialchars($search) ?>" class="search-input">
        <select name="department" class="filter-select">
          <option value="0">All Academic Departments</option>
          <?php foreach ($departments as $d): ?>
            <option value="<?= $d['dept_id'] ?>" <?= $deptFilter == $d['dept_id'] ? 'selected' : '' ?>>
              <?= htmlspecialchars($d['dept_code']) ?> - <?= htmlspecialchars($d['dept_name']) ?>
```

```
          </option>
        <?php endforeach; ?>
      </select>
      <button type="submit" class="btn btn-secondary">Filter Records</button>
      <?php if (!empty($search) || $deptFilter > 0): ?>
        <a href="index.php" class="btn btn-outline">Reset Filter</a>
      <?php endif; ?>
    </form>
  </div>

  <!-- Student Records Table -->
  <div class="table-card">
    <table class="data-table">
      <thead>
        <tr>
          <th>Photo</th>
          <th>Roll Number</th>
          <th>Candidate Full Name</th>
          <th>Department</th>
          <th>CGPA</th>
          <th>Contact Info</th>
          <th>Status</th>
          <th>Document</th>
          <th>Actions</th>
        </tr>
      </thead>
      <tbody>
        <?php if (empty($students)): ?>
          <tr>
            <td colspan="9" class="empty-state">No matching student records located in the institutional database.</td>
          </tr>
        <?php else: ?>
          <?php foreach ($students as $row): ?>
            <tr>
              <td>
                <?php if (!empty($row['photo_path']) && file_exists($row['photo_path'])): ?>
                  <img src="<?= htmlspecialchars($row['photo_path']) ?>" class="avatar-thumb" alt="Avatar">
                <?php else: ?>
                  <div class="avatar-placeholder"><?= strtoupper(substr($row['first_name'], 0, 1)) ?></div>
                <?php endif; ?>
              </td>
              <td><strong><?= htmlspecialchars($row['roll_no']) ?></strong></td>
              <td><?= htmlspecialchars($row['first_name'] . ' ' . $row['last_name']) ?></td>
              <td><span class="dept-badge"><?= htmlspecialchars($row['dept_code']) ?></span></td>
              <td><strong><?= number_format($row['cgpa'], 2) ?></strong></td>
              <td class="contact-cell">
                <div><?= htmlspecialchars($row['email']) ?></div>
                <div class="text-muted">+91 <?= htmlspecialchars($row['phone']) ?></div>
              </td>
              <td>
                <span class="status-badge status-<?= strtolower($row['enrollment_status']) ?>">
                  <?= htmlspecialchars($row['enrollment_status']) ?>
                </span>
              </td>
              <td>
                <?php if (!empty($row['doc_path']) && file_exists($row['doc_path'])): ?>
                  <a href="<?= htmlspecialchars($row['doc_path']) ?>" target="_blank" class="doc-link" title="View
Document">&#x1F4C4; PDF</a>
                <?php else: ?>
```

```
                <span class="text-muted">None</span>
              <?php endif; ?>
            </td>
            <td class="action-cell">
              <a href="view_student.php?id=<?= $row['student_id'] ?>" class="btn-action btn-view" title="View
Profile">&#x1F441;</a>
              <a href="edit_student.php?id=<?= $row['student_id'] ?>" class="btn-action btn-edit" title="Edit
Record">&#x270E;</a>
              <a href="delete_student.php?id=<?= $row['student_id'] ?>" class="btn-action btn-delete" onclick="return
confirm('Permanently remove student record and unlinked disk files?');" title="Delete">&#x1F5D1;</a>
            </td>
          </tr>
        <?php endforeach; ?>
      <?php endif; ?>
    </tbody>
  </table>
 </div>
</div>
</body>
</html>
```

add_student.php:

```
<?php
// ============================================================================
// Student Enrollment Controller (add_student.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// ============================================================================
require_once 'db.php';

$errors = [];
$departments = $pdo->query("SELECT * FROM departments ORDER BY dept_code ASC")->fetchAll();

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $rollNo = strtoupper(trim($_POST['roll_no'] ?? ''));
    $firstName = trim($_POST['first_name'] ?? '');
    $lastName = trim($_POST['last_name'] ?? '');
    $email = trim($_POST['email'] ?? '');
    $phone = trim($_POST['phone'] ?? '');
    $gender = $_POST['gender'] ?? 'Male';
    $dob = $_POST['dob'] ?? '';
    $deptId = intval($_POST['dept_id'] ?? 0);
    $cgpa = floatval($_POST['cgpa'] ?? 0.0);
    $status = $_POST['enrollment_status'] ?? 'Active';

    // Server-Side Validations
    if (empty($rollNo)) $errors[] = "Roll Number is required.";
    if (empty($firstName) || empty($lastName)) $errors[] = "First and Last names are mandatory.";
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = "Valid email address is required.";
    if (!preg_match('/^[6-9][0-9]{9}$/', $phone)) $errors[] = "Phone must be a valid 10-digit mobile number.";
    if ($deptId <= 0) $errors[] = "Please select a valid academic department.";
    if ($cgpa < 0.0 || $cgpa > 10.0) $errors[] = "CGPA must be between 0.00 and 10.00.";

    // Uniqueness Verification
    $checkStmt = $pdo->prepare("SELECT COUNT(*) FROM students WHERE roll_no = :r OR email = :e");
    $checkStmt->execute([':r' => $rollNo, ':e' => $email]);
    if ($checkStmt->fetchColumn() > 0) {
        $errors[] = "Roll Number ({$rollNo}) or Email ({$email}) is already registered!";
    }
```

```
    // Process Profile Photo Upload
    $photoPath = null;
    if (isset($_FILES['photo']) && $_FILES['photo']['error'] === UPLOAD_ERR_OK) {
        $finfo = finfo_open(FILEINFO_MIME_TYPE);
        $mime = finfo_file($finfo, $_FILES['photo']['tmp_name']);
        finfo_close($finfo);

        $allowedMimes = ['image/jpeg', 'image/png', 'image/jpg'];
        if (!in_array($mime, $allowedMimes)) {
            $errors[] = "Invalid photo format. Only JPEG and PNG are allowed.";
        } elseif ($_FILES['photo']['size'] > 2 * 1024 * 1024) {
            $errors[] = "Photo size must not exceed 2MB.";
        } else {
            $ext = ($mime === 'image/png') ? 'png' : 'jpg';
            $uniquePhotoName = 'avatar_' . bin2hex(random_bytes(6)) . '_' . time() . '.' . $ext;
            $photoPath = 'uploads/photos/' . $uniquePhotoName;
            move_uploaded_file($_FILES['photo']['tmp_name'], PHOTO_DIR . $uniquePhotoName);
        }
    }

    // Process Document Upload (PDF)
    $docPath = null;
    $docOrigName = null;
    if (isset($_FILES['id_doc']) && $_FILES['id_doc']['error'] === UPLOAD_ERR_OK) {
        $finfo = finfo_open(FILEINFO_MIME_TYPE);
        $mime = finfo_file($finfo, $_FILES['id_doc']['tmp_name']);
        finfo_close($finfo);

        if ($mime !== 'application/pdf') {
            $errors[] = "Official ID document must be a verified PDF file.";
        } elseif ($_FILES['id_doc']['size'] > 5 * 1024 * 1024) {
            $errors[] = "Document size exceeds 5MB limit.";
        } else {
            $docOrigName = basename($_FILES['id_doc']['name']);
            $uniqueDocName = 'doc_' . bin2hex(random_bytes(6)) . '_' . time() . '.pdf';
            $docPath = 'uploads/documents/' . $uniqueDocName;
            move_uploaded_file($_FILES['id_doc']['tmp_name'], DOC_DIR . $uniqueDocName);
        }
    }

    // Insert into Database if valid
    if (empty($errors)) {
        try {
            $pdo->beginTransaction();
            $sql = "INSERT INTO students (roll_no, first_name, last_name, email, phone, gender, dob, dept_id, cgpa,
photo_path, doc_path, doc_original_name, enrollment_status)
                    VALUES (:roll_no, :first_name, :last_name, :email, :phone, :gender, :dob, :dept_id, :cgpa, :photo_path,
:doc_path, :doc_orig, :status)";
            $stmt = $pdo->prepare($sql);
            $stmt->execute([
                ':roll_no'    => $rollNo,
                ':first_name' => $firstName,
                ':last_name'  => $lastName,
                ':email'      => $email,
                ':phone'      => $phone,
                ':gender'     => $gender,
                ':dob'        => $dob,
                ':dept_id'    => $deptId,
                ':cgpa'       => $cgpa,
```

```
            ':photo_path' => $photoPath,
            ':doc_path'   => $docPath,
            ':doc_orig'   => $docOrigName,
            ':status'     => $status
          ]);
          $pdo->commit();
          setFlash('success', "Student record [{$rollNo}] {$firstName} {$lastName} successfully enrolled.");
          header("Location: index.php");
          exit;
        } catch (PDOException $e) {
          $pdo->rollBack();
          $errors[] = "Database Insertion Failed: " . $e->getMessage();
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Enroll New Student - Apex College Administration</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="form-container">
    <div class="card-box">
      <h2>&#x1F4DD; Student Enrollment &amp;amp; Document Registration</h2>
      <p class="subtitle">Complete the fields below to register a scholar and upload attachments.</p>

      <?php if (!empty($errors)): ?>
        <div class="alert alert-danger">
          <ul>
            <?php foreach ($errors as $err): ?>
              <li><?= htmlspecialchars($err) ?></li>
            <?php endforeach; ?>
          </ul>
        </div>
      <?php endif; ?>

      <form method="POST" action="add_student.php" enctype="multipart/form-data" class="portal-form">
        <div class="form-row">
          <div class="form-col">
            <label>Roll Number *</label>
            <input type="text" name="roll_no" placeholder="e.g. 24CS039" value="<?= htmlspecialchars($_POST['roll_no'] ??
") ?>" required>
          </div>
          <div class="form-col">
            <label>Academic Department *</label>
            <select name="dept_id" required>
              <option value="">-- Choose Department --</option>
              <?php foreach ($departments as $d): ?>
                <option value="<?= $d['dept_id'] ?>" <?= (isset($_POST['dept_id']) && $_POST['dept_id'] == $d['dept_id']) ?
'selected' : '' ?>>
                  <?= htmlspecialchars($d['dept_code']) ?> - <?= htmlspecialchars($d['dept_name']) ?>
                </option>
              <?php endforeach; ?>
            </select>
          </div>
        </div>
```

```
      <div class="form-row">
        <div class="form-col">
          <label>First Name *</label>
          <input type="text" name="first_name" placeholder="Given Name" value="<?=
htmlspecialchars($_POST['first_name'] ?? '') ?>" required>
        </div>
        <div class="form-col">
          <label>Last Name *</label>
          <input type="text" name="last_name" placeholder="Surname / Initial" value="<?=
htmlspecialchars($_POST['last_name'] ?? '') ?>" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Email Address *</label>
          <input type="email" name="email" placeholder="official@apex-tech.edu.in" value="<?=
htmlspecialchars($_POST['email'] ?? '') ?>" required>
        </div>
        <div class="form-col">
          <label>Phone Number *</label>
          <input type="tel" name="phone" placeholder="10-digit mobile" value="<?= htmlspecialchars($_POST['phone'] ?? '')
?>" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Date of Birth *</label>
          <input type="date" name="dob" value="<?= htmlspecialchars($_POST['dob'] ?? '') ?>" required>
        </div>
        <div class="form-col">
          <label>Cumulative GPA (CGPA) *</label>
          <input type="number" step="0.01" min="0" max="10" name="cgpa" placeholder="e.g. 9.45" value="<?=
htmlspecialchars($_POST['cgpa'] ?? '') ?>" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Gender *</label>
          <select name="gender">
            <option value="Male">Male</option>
            <option value="Female">Female</option>
            <option value="Other">Other</option>
          </select>
        </div>
        <div class="form-col">
          <label>Enrollment Status *</label>
          <select name="enrollment_status">
            <option value="Active">Active</option>
            <option value="Graduated">Graduated</option>
            <option value="Suspended">Suspended</option>
          </select>
        </div>
      </div>

      <div class="form-row">
```

```
        <div class="form-col">
          <label>Student Photo (JPEG / PNG, Max 2MB)</label>
          <input type="file" name="photo" accept="image/jpeg,image/png">
        </div>
        <div class="form-col">
          <label>Identification Document (PDF only, Max 5MB)</label>
          <input type="file" name="id_doc" accept="application/pdf">
        </div>
      </div>

      <div class="form-footer">
        <a href="index.php" class="btn btn-outline">&larr; Return to Dashboard</a>
        <button type="submit" class="btn btn-primary">&#x2714; Save &amp;amp; Enroll Scholar</button>
      </div>
    </form>
  </div>
 </div>
</body>
</html>
```

view_student.php:

```
<?php
// ============================================================================
// Student Profile Dossier Viewer (view_student.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// ============================================================================
require_once 'db.php';

$id = intval($_GET['id'] ?? 0);
$stmt = $pdo->prepare("SELECT s.*, d.dept_code, d.dept_name, d.hod_name
                       FROM students s
                       JOIN departments d ON s.dept_id = d.dept_id
                       WHERE s.student_id = :id");
$stmt->execute([':id' => $id]);
$student = $stmt->fetch();

if (!$student) {
    setFlash('danger', "Student dossier not found.");
    header("Location: index.php");
    exit;
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>Student Dossier - <?= htmlspecialchars($student['roll_no']) ?></title>
 <link rel="stylesheet" href="style.css">
</head>
<body>
 <div class="dossier-wrapper">
  <div class="card-box">
   <div class="dossier-header">
    <div class="dossier-photo-container">
     <?php if (!empty($student['photo_path']) && file_exists($student['photo_path'])): ?>
      <img src="<?= htmlspecialchars($student['photo_path']) ?>" class="dossier-photo" alt="Photo">
     <?php else: ?>
      <div class="dossier-placeholder"><?= strtoupper(substr($student['first_name'], 0, 1)) ?></div>
     <?php endif; ?>
```

```
      </div>
      <div class="dossier-intro">
        <h2><?= htmlspecialchars($student['first_name'] . ' ' . $student['last_name']) ?></h2>
        <p class="dossier-meta">
          <strong>Roll No:</strong> <?= htmlspecialchars($student['roll_no']) ?> &bull;
          <strong>Status:</strong> <span class="status-badge status-<?= strtolower($student['enrollment_status']) ?>"><?=
htmlspecialchars($student['enrollment_status']) ?></span>
        </p>
        <p class="dossier-dept"><?= htmlspecialchars($student['dept_name']) ?> (<?= htmlspecialchars($student['dept_code'])
?>)</p>
      </div>
    </div>

    <div class="dossier-grid">
      <div class="dossier-item">
        <span class="label">Head of Department:</span>
        <span class="val"><?= htmlspecialchars($student['hod_name']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Cumulative GPA:</span>
        <span class="val"><strong><?= number_format($student['cgpa'], 2) ?> / 10.00</strong></span>
      </div>
      <div class="dossier-item">
        <span class="label">Institutional Email:</span>
        <span class="val"><?= htmlspecialchars($student['email']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Phone Contact:</span>
        <span class="val">+91 <?= htmlspecialchars($student['phone']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Date of Birth:</span>
        <span class="val"><?= htmlspecialchars($student['dob']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Gender:</span>
        <span class="val"><?= htmlspecialchars($student['gender']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Date Registered:</span>
        <span class="val"><?= htmlspecialchars($student['created_at']) ?></span>
      </div>
      <div class="dossier-item">
        <span class="label">Last Record Update:</span>
        <span class="val"><?= htmlspecialchars($student['updated_at']) ?></span>
      </div>
    </div>

    <div class="dossier-document-section">
      <h3>Attached Verification Document</h3>
      <?php if (!empty($student['doc_path']) && file_exists($student['doc_path'])): ?>
        <div class="doc-box">
          <span class="doc-icon">&#x1F4C4;</span>
          <div class="doc-info">
            <span class="doc-name"><?= htmlspecialchars($student['doc_original_name'] ?: 'Identification_Certificate.pdf')
?></span>
            <span class="doc-size"><?= round(filesize($student['doc_path']) / 1024, 2) ?> KB</span>
          </div>
```

```
          <a href="<?= htmlspecialchars($student['doc_path']) ?>" target="_blank" class="btn btn-secondary">&#x2913;
Download &amp;amp; Inspect</a>
        </div>
       <?php else: ?>
        <p class="text-muted">No identification document on server storage for this scholar.</p>
       <?php endif; ?>
      </div>

      <div class="form-footer">
       <a href="index.php" class="btn btn-outline">&larr; Return to Dashboard</a>
       <a href="edit_student.php?id=<?= $student['student_id'] ?>" class="btn btn-primary">&#x270E; Edit Dossier</a>
      </div>
    </div>
  </div>
</body>
</html>
```

edit_student.php:

```
<?php
// ============================================================================
// Student Record Modification Controller (edit_student.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// ============================================================================
require_once 'db.php';

$id = intval($_GET['id'] ?? 0);
$stmt = $pdo->prepare("SELECT * FROM students WHERE student_id = :id");
$stmt->execute([':id' => $id]);
$student = $stmt->fetch();

if (!$student) {
    setFlash('danger', "Student record not found for modification.");
    header("Location: index.php");
    exit;
}

$departments = $pdo->query("SELECT * FROM departments ORDER BY dept_code ASC")->fetchAll();
$errors = [];

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $firstName = trim($_POST['first_name'] ?? '');
    $lastName = trim($_POST['last_name'] ?? '');
    $email = trim($_POST['email'] ?? '');
    $phone = trim($_POST['phone'] ?? '');
    $gender = $_POST['gender'] ?? 'Male';
    $dob = $_POST['dob'] ?? '';
    $deptId = intval($_POST['dept_id'] ?? 0);
    $cgpa = floatval($_POST['cgpa'] ?? 0.0);
    $status = $_POST['enrollment_status'] ?? 'Active';

    if (empty($firstName) || empty($lastName)) $errors[] = "Names cannot be empty.";
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = "Valid email required.";
    if (!preg_match('/^[6-9][0-9]{9}$/', $phone)) $errors[] = "Valid 10-digit phone required.";
    if ($cgpa < 0.0 || $cgpa > 10.0) $errors[] = "CGPA must be between 0.00 and 10.00.";

    // Check email uniqueness excluding current student
    $dupStmt = $pdo->prepare("SELECT COUNT(*) FROM students WHERE email = :e AND student_id != :id");
    $dupStmt->execute([':e' => $email, ':id' => $id]);
    if ($dupStmt->fetchColumn() > 0) {
```

```
    $errors[] = "Email address is already in use by another student.";
}

$photoPath = $student['photo_path'];
if (isset($_FILES['photo']) && $_FILES['photo']['error'] === UPLOAD_ERR_OK) {
    $finfo = finfo_open(FILEINFO_MIME_TYPE);
    $mime = finfo_file($finfo, $_FILES['photo']['tmp_name']);
    finfo_close($finfo);

    $allowedMimes = ['image/jpeg', 'image/png', 'image/jpg'];
    if (in_array($mime, $allowedMimes)) {
        // Unlink old photo if exists
        if (!empty($photoPath) && file_exists($photoPath)) {
            unlink($photoPath);
        }
        $ext = ($mime === 'image/png') ? 'png' : 'jpg';
        $uniquePhotoName = 'avatar_' . bin2hex(random_bytes(6)) . '_' . time() . '.' . $ext;
        $photoPath = 'uploads/photos/' . $uniquePhotoName;
        move_uploaded_file($_FILES['photo']['tmp_name'], PHOTO_DIR . $uniquePhotoName);
    } else {
        $errors[] = "Invalid photo format. Only JPG/PNG permitted.";
    }
}

$docPath = $student['doc_path'];
$docOrigName = $student['doc_original_name'];
if (isset($_FILES['id_doc']) && $_FILES['id_doc']['error'] === UPLOAD_ERR_OK) {
    $finfo = finfo_open(FILEINFO_MIME_TYPE);
    $mime = finfo_file($finfo, $_FILES['id_doc']['tmp_name']);
    finfo_close($finfo);

    if ($mime === 'application/pdf') {
        // Unlink old document if exists
        if (!empty($docPath) && file_exists($docPath)) {
            unlink($docPath);
        }
        $docOrigName = basename($_FILES['id_doc']['name']);
        $uniqueDocName = 'doc_' . bin2hex(random_bytes(6)) . '_' . time() . '.pdf';
        $docPath = 'uploads/documents/' . $uniqueDocName;
        move_uploaded_file($_FILES['id_doc']['tmp_name'], DOC_DIR . $uniqueDocName);
    } else {
        $errors[] = "ID Document must be a valid PDF.";
    }
}

if (empty($errors)) {
    $updateSql = "UPDATE students SET
                  first_name = :first_name,
                  last_name = :last_name,
                  email = :email,
                  phone = :phone,
                  gender = :gender,
                  dob = :dob,
                  dept_id = :dept_id,
                  cgpa = :cgpa,
                  photo_path = :photo_path,
                  doc_path = :doc_path,
                  doc_original_name = :doc_orig,
                  enrollment_status = :status
```

```
                WHERE student_id = :id";
        $updateStmt = $pdo->prepare($updateSql);
        $updateStmt->execute([
            ':first_name' => $firstName,
            ':last_name'  => $lastName,
            ':email'      => $email,
            ':phone'      => $phone,
            ':gender'     => $gender,
            ':dob'        => $dob,
            ':dept_id'    => $deptId,
            ':cgpa'       => $cgpa,
            ':photo_path' => $photoPath,
            ':doc_path'   => $docPath,
            ':doc_orig'   => $docOrigName,
            ':status'     => $status,
            ':id'         => $id
        ]);

        setFlash('success', "Student record [{$student['roll_no']}] successfully updated.");
        header("Location: index.php");
        exit;
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Edit Student - <?= htmlspecialchars($student['roll_no']) ?></title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="form-container">
    <div class="card-box">
      <h2>&#x270E; Modify Student Information - <?= htmlspecialchars($student['roll_no']) ?></h2>

      <?php if (!empty($errors)): ?>
        <div class="alert alert-danger">
          <ul><?php foreach ($errors as $e): ?><li><?= htmlspecialchars($e) ?></li><?php endforeach; ?></ul>
        </div>
      <?php endif; ?>

      <form method="POST" action="edit_student.php?id=<?= $id ?>" enctype="multipart/form-data" class="portal-form">
        <div class="form-row">
          <div class="form-col">
            <label>Roll Number (Permanent Identifier)</label>
            <input type="text" value="<?= htmlspecialchars($student['roll_no']) ?>" disabled class="input-disabled">
          </div>
          <div class="form-col">
            <label>Academic Department</label>
            <select name="dept_id">
              <?php foreach ($departments as $d): ?>
                <option value="<?= $d['dept_id'] ?>" <?= $student['dept_id'] == $d['dept_id'] ? 'selected' : '' ?>>
                  <?= htmlspecialchars($d['dept_code']) ?> - <?= htmlspecialchars($d['dept_name']) ?>
                </option>
              <?php endforeach; ?>
            </select>
          </div>
        </div>
```

```
      <div class="form-row">
        <div class="form-col">
          <label>First Name</label>
          <input type="text" name="first_name" value="<?= htmlspecialchars($student['first_name']) ?>" required>
        </div>
        <div class="form-col">
          <label>Last Name</label>
          <input type="text" name="last_name" value="<?= htmlspecialchars($student['last_name']) ?>" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Email Address</label>
          <input type="email" name="email" value="<?= htmlspecialchars($student['email']) ?>" required>
        </div>
        <div class="form-col">
          <label>Phone Number</label>
          <input type="tel" name="phone" value="<?= htmlspecialchars($student['phone']) ?>" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Cumulative GPA</label>
          <input type="number" step="0.01" min="0" max="10" name="cgpa" value="<?= htmlspecialchars($student['cgpa'])
?>" required>
        </div>
        <div class="form-col">
          <label>Enrollment Status</label>
          <select name="enrollment_status">
            <option value="Active" <?= $student['enrollment_status'] === 'Active' ? 'selected' : '' ?>>Active</option>
            <option value="Graduated" <?= $student['enrollment_status'] === 'Graduated' ? 'selected' : ''
?>>Graduated</option>
            <option value="Suspended" <?= $student['enrollment_status'] === 'Suspended' ? 'selected' : ''
?>>Suspended</option>
          </select>
        </div>
      </div>

      <div class="form-row">
        <div class="form-col">
          <label>Replace Photograph (Leave blank to keep current)</label>
          <input type="file" name="photo" accept="image/jpeg,image/png">
        </div>
        <div class="form-col">
          <label>Replace ID Document (PDF, Leave blank to retain)</label>
          <input type="file" name="id_doc" accept="application/pdf">
        </div>
      </div>

      <div class="form-footer">
        <a href="index.php" class="btn btn-outline">&larr; Cancel</a>
        <button type="submit" class="btn btn-primary">&#x2714; Update Student Details</button>
      </div>
    </form>
  </div>
</div>
```

```
</body>
</html>
```

delete_student.php:

```
<?php
// ========================================================================
// Student Record Deletion Controller (delete_student.php)
// Student: DARSHAN SUNDAR K K | Roll No: 24CS039
// Unlinks physical disk files and removes MySQL database row
// ========================================================================
require_once 'db.php';

$id = intval($_GET['id'] ?? 0);

if ($id <= 0) {
    setFlash('danger', "Invalid student identifier provided for deletion.");
    header("Location: index.php");
    exit;
}

try {
    $pdo->beginTransaction();

    // Retrieve file paths prior to deletion
    $stmt = $pdo->prepare("SELECT roll_no, first_name, last_name, photo_path, doc_path FROM students WHERE
student_id = :id");
    $stmt->execute([':id' => $id]);
    $student = $stmt->fetch();

    if ($student) {
        // Physical file unlinking from server storage
        if (!empty($student['photo_path']) && file_exists($student['photo_path'])) {
            unlink($student['photo_path']);
        }
        if (!empty($student['doc_path']) && file_exists($student['doc_path'])) {
            unlink($student['doc_path']);
        }

        // Delete row from MySQL
        $delStmt = $pdo->prepare("DELETE FROM students WHERE student_id = :id");
        $delStmt->execute([':id' => $id]);

        $pdo->commit();
        setFlash('success', "Student record [{$student['roll_no']}] {$student['first_name']} and associated files were permanently
deleted.");
    } else {
        $pdo->rollBack();
        setFlash('danger', "Target student record does not exist.");
    }
} catch (PDOException $e) {
    $pdo->rollBack();
    setFlash('danger', "Deletion failed: " . $e->getMessage());
}

header("Location: index.php");
exit;
?>
```

style.css:

```
/* Apex College Administration Portal Stylesheet */
```

```
:root {
  --navy-primary: #0f172a;
  --navy-secondary: #1e293b;
  --accent-amber: #d97706;
  --bg-slate: #f8fafc;
  --card-bg: #ffffff;
  --border-color: #e2e8f0;
  --text-main: #1e293b;
  --text-muted: #64748b;
  --color-success: #059669;
  --color-danger: #dc2626;
  --font-family: 'Times New Roman', Times, serif;
}

* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: var(--font-family);
  background-color: var(--bg-slate);
  color: var(--text-main);
  line-height: 1.45;
  padding: 1.5rem;
}

.dashboard-wrapper, .form-container, .dossier-wrapper {
  max-width: 1200px;
  margin: 0 auto;
}

.top-bar {
  background-color: var(--navy-primary);
  color: #ffffff;
  padding: 1.25rem 1.75rem;
  border-radius: 8px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1.5rem;
  border-bottom: 3px solid var(--accent-amber);
}

.top-bar h1 {
  font-size: 1.35rem;
  letter-spacing: 0.5px;
}

.tagline {
  font-size: 0.85rem;
  color: #cbd5e1;
}

.kpi-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 1rem;
```

```
  margin-bottom: 1.5rem;
}

.kpi-card {
  background-color: var(--card-bg);
  padding: 1.25rem;
  border-radius: 8px;
  border: 1px solid var(--border-color);
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.05);
  display: flex;
  flex-direction: column;
}

.kpi-title {
  font-size: 0.82rem;
  font-weight: bold;
  color: var(--text-muted);
  text-transform: uppercase;
}

.kpi-value {
  font-size: 1.8rem;
  font-weight: bold;
  color: var(--navy-primary);
  margin-top: 0.35rem;
}

.filter-card {
  background-color: var(--card-bg);
  padding: 1rem 1.25rem;
  border-radius: 8px;
  border: 1px solid var(--border-color);
  margin-bottom: 1.25rem;
}

.filter-form {
  display: flex;
  gap: 0.75rem;
  flex-wrap: wrap;
}

.search-input {
  flex: 1;
  min-width: 280px;
  padding: 0.55rem 0.85rem;
  border: 1.5px solid var(--border-color);
  border-radius: 6px;
  font-family: var(--font-family);
  font-size: 0.92rem;
}

.filter-select {
  padding: 0.55rem 0.85rem;
  border: 1.5px solid var(--border-color);
  border-radius: 6px;
  font-family: var(--font-family);
  font-size: 0.92rem;
}
```

```
.table-card {
  background-color: var(--card-bg);
  border-radius: 8px;
  border: 1px solid var(--border-color);
  overflow-x: auto;
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.06);
}

.data-table {
  width: 100%;
  border-collapse: collapse;
  text-align: left;
}

.data-table th {
  background-color: #f1f5f9;
  padding: 0.75rem 1rem;
  font-size: 0.88rem;
  color: var(--navy-secondary);
  border-bottom: 2px solid var(--border-color);
}

.data-table td {
  padding: 0.75rem 1rem;
  font-size: 0.9rem;
  border-bottom: 1px solid var(--border-color);
  vertical-align: middle;
}

.avatar-thumb {
  width: 38px;
  height: 38px;
  border-radius: 50%;
  object-fit: cover;
  border: 1.5px solid var(--border-color);
}

.avatar-placeholder {
  width: 38px;
  height: 38px;
  border-radius: 50%;
  background-color: #e2e8f0;
  color: var(--navy-primary);
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: bold;
}

.dept-badge {
  background-color: #e0f2fe;
  color: #0369a1;
  padding: 0.2rem 0.55rem;
  border-radius: 4px;
  font-size: 0.82rem;
  font-weight: bold;
}

.status-badge {
```

```
  padding: 0.2rem 0.55rem;
  border-radius: 4px;
  font-size: 0.8rem;
  font-weight: bold;
}

.status-active {
  background-color: #ecfdf5;
  color: #047857;
}

.status-graduated {
  background-color: #f0f9ff;
  color: #0284c7;
}

.status-suspended {
  background-color: #fef2f2;
  color: #b91c1c;
}

.doc-link {
  color: var(--accent-amber);
  font-weight: bold;
  text-decoration: none;
}

.action-cell {
  white-space: nowrap;
}

.btn-action {
  display: inline-block;
  padding: 0.3rem 0.55rem;
  border-radius: 4px;
  text-decoration: none;
  font-size: 0.85rem;
  margin-right: 0.25rem;
}

.btn-view { background-color: #e2e8f0; color: #1e293b; }
.btn-edit { background-color: #fef3c7; color: #92400e; }
.btn-delete { background-color: #fee2e2; color: #991b1b; }

.btn {
  font-family: var(--font-family);
  font-size: 0.9rem;
  font-weight: bold;
  padding: 0.55rem 1.1rem;
  border-radius: 6px;
  cursor: pointer;
  border: none;
  text-decoration: none;
  display: inline-block;
}

.btn-primary { background-color: var(--navy-primary); color: #ffffff; }
.btn-secondary { background-color: #e2e8f0; color: var(--text-main); }
.btn-outline { background-color: transparent; border: 1.5px solid var(--border-color); color: var(--text-main); }
```

```
.alert {
  padding: 0.85rem 1.25rem;
  border-radius: 6px;
  margin-bottom: 1.25rem;
  font-size: 0.92rem;
}

.alert-success { background-color: #ecfdf5; color: #065f46; border: 1px solid #a7f3d0; }
.alert-danger { background-color: #fef2f2; color: #991b1b; border: 1px solid #fecaca; }

.card-box {
  background: #ffffff;
  padding: 1.75rem;
  border-radius: 8px;
  border: 1px solid var(--border-color);
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.06);
}

.portal-form .form-row {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 1.2rem;
  margin-bottom: 1rem;
}

.form-col {
  display: flex;
  flex-direction: column;
}

.form-col label {
  font-size: 0.88rem;
  font-weight: bold;
  margin-bottom: 0.35rem;
}

.form-col input, .form-col select {
  font-family: var(--font-family);
  font-size: 0.9rem;
  padding: 0.55rem 0.75rem;
  border: 1.5px solid var(--border-color);
  border-radius: 5px;
}

.form-footer {
  display: flex;
  justify-content: flex-end;
  gap: 0.75rem;
  margin-top: 1.5rem;
  padding-top: 1.25rem;
  border-top: 1px solid var(--border-color);
}

.dossier-header {
  display: flex;
  gap: 1.5rem;
  align-items: center;
  border-bottom: 1px solid var(--border-color);
```

```
  padding-bottom: 1.5rem;
  margin-bottom: 1.5rem;
}

.dossier-photo {
  width: 90px;
  height: 90px;
  border-radius: 8px;
  object-fit: cover;
  border: 2px solid var(--border-color);
}

.dossier-placeholder {
  width: 90px;
  height: 90px;
  border-radius: 8px;
  background-color: #e2e8f0;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 2.2rem;
  font-weight: bold;
}

.dossier-grid {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 1rem;
  margin-bottom: 1.5rem;
}

.dossier-item {
  display: flex;
  flex-direction: column;
}

.dossier-item .label {
  font-size: 0.82rem;
  color: var(--text-muted);
}

.dossier-item .val {
  font-size: 0.98rem;
}

.dossier-document-section {
  background-color: #f8fafc;
  padding: 1.25rem;
  border-radius: 6px;
  border: 1px dashed var(--border-color);
}

.doc-box {
  display: flex;
  align-items: center;
  gap: 1rem;
  margin-top: 0.75rem;
}
```

```
.doc-icon {
  font-size: 2rem;
}

.doc-info {
  flex: 1;
  display: flex;
  flex-direction: column;
}

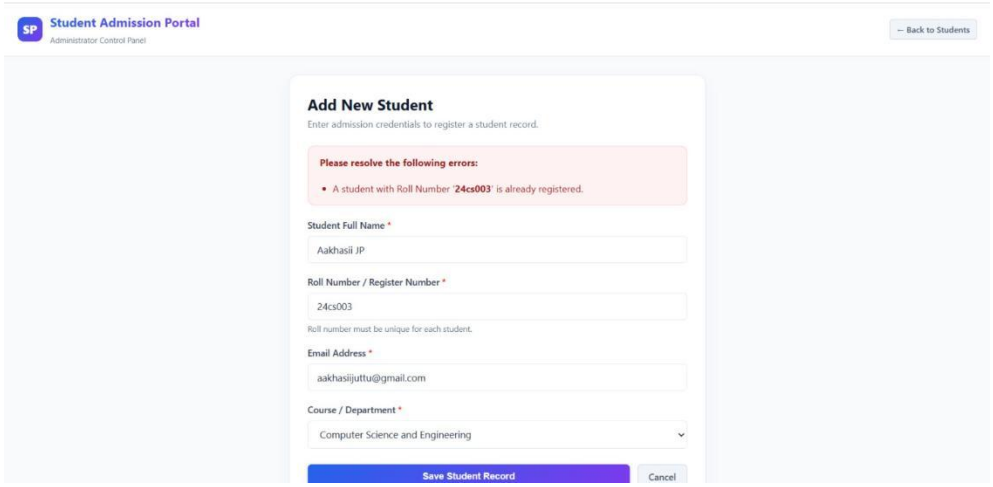
.doc-name {
  font-weight: bold;
}

.doc-size {
  font-size: 0.8rem;
  color: var(--text-muted);
}
```

OUTPUT :

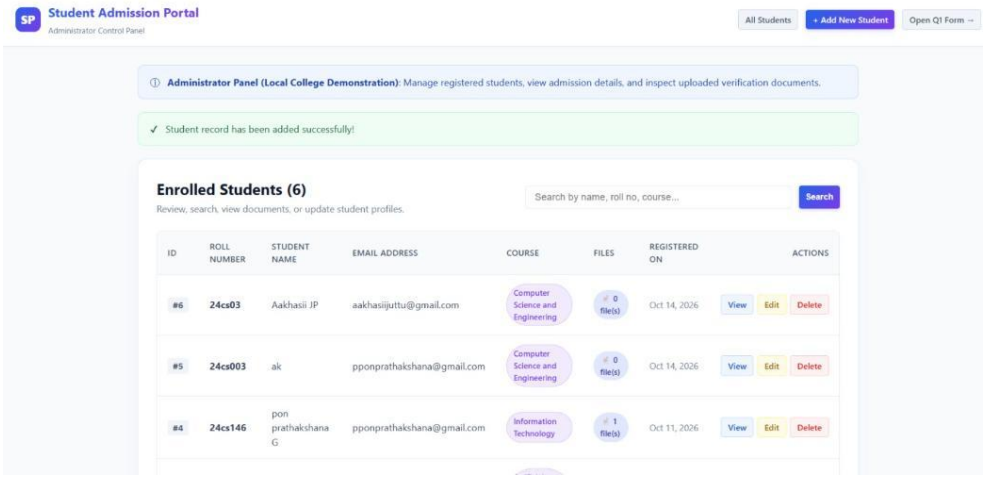
INVALID OPERATIONS :

Invalid Operation – Roll Number Already Exists :

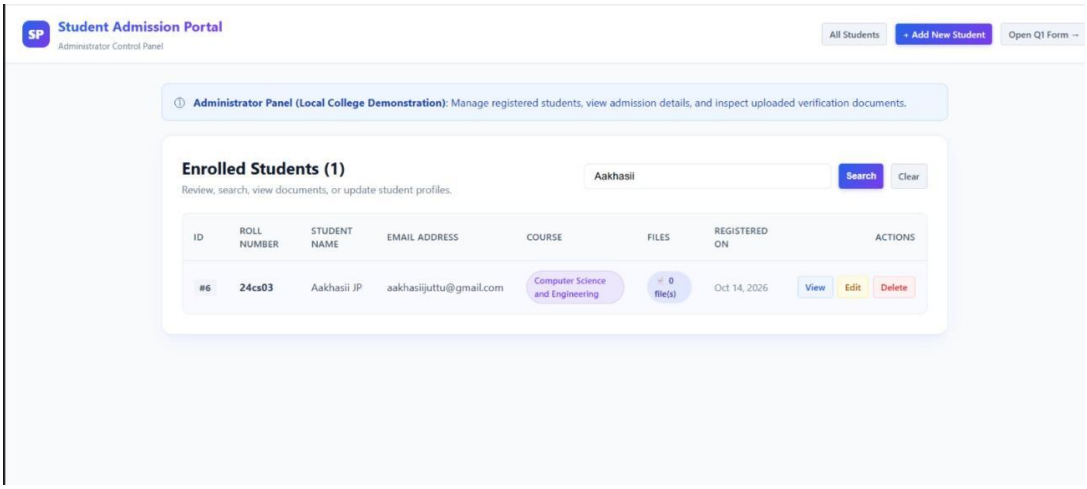


VALID OPERATIONS :

Valid Operation – Student Records Directory & Action Management (CRUD Read) :



Valid Operation – Search & Filter Student Records by Name :



RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Relational Database Schema & Integrity (R1)	4	
Complete CRUD Implementation (R2)	4	
File Upload & MIME Verification (R3)	4	
Storage Security & File Unlink Management (R4)	4	
UI Presentation & Flash Messaging (R5)	4	
Total Marks	**20**	

RESULT:

A full-stack College Administration Panel for student records and file attachments was successfully implemented using PHP and MySQL. The system executes parameterized CRUD transactions, strictly verifies file size and MIME properties, assigns hashed collision-free names, tracks comprehensive document metadata, and maintains synchronized physical storage by unlinking files during record deletions.

Q No: 3	Dynamic student profiles portal with image & document streaming using Java Servlets, JDBC & JSP

AIM:

To develop an enterprise-grade Java Web Application implementing the Model-View-Controller (MVC) architecture using Java Servlets, Java Database Connectivity (JDBC), and JavaServer Pages (JSP). The application enforces stateful administrative session authentication, executes parameterized JDBC queries over MySQL database tables to retrieve scholar records, streams binary photo assets directly to client <img> elements via dedicated servlet endpoints using Buffered Streams and dynamic MIME headers, conditionally renders document download links for verified files, and provides robust session invalidation upon administrative logout.

PROCEDURE:

1. Define the Java web deployment descriptor ('web.xml') establishing servlet definitions, URL mappings, session timeout thresholds, and default welcome file navigation.
2. Author a thread-safe singleton JDBC connector ('DBConfig.java') to manage MySQL driver initialization, database connection parameters, and structured resource releases.
3. Design the administrative authentication controller ('AdminAuthServlet.java') verifying credentials against secure parameters and provisioning stateful HttpSession attributes.
4. Construct the administrative termination controller ('PortalLogoutServlet.java') to invalidate active sessions, flush response caches, and route users back to login with status messaging.
5. Develop the core data retrieval controller ('StudentDirectoryServlet.java') querying student profiles joined with department entities, encapsulating results into request-scoped collections.
6. Architect the high-performance binary streaming engine ('MediaStreamServlet.java') reading image and document byte streams from protected server disk paths directly to HttpServletResponse output streams.
7. Implement strict HTTP header management in the streaming servlet, inspecting file extensions to dynamically assign appropriate Content-Type ('image/jpeg', 'image/png', 'application/pdf') and Content-Disposition headers.
8. Author the administrator login interface ('login.jsp') featuring form input validation, contextual error alert rendering, and professional institutional styling.
9. Design the dynamic student profiles view ('students.jsp') rendering responsive profile cards, embedding byte-streamed photos via <img src='mediaStream?id=...&type=photo'>, and displaying conditional document access buttons.
10. Create an elegant, unified CSS stylesheet ('style.css') delivering card-based visual hierarchy, status badges, responsive grid layouts, and smooth hover transitions.

SERVLET & JSP CONSTRUCTS USED:

@WebServlet / web.xml: Servlet configuration and declarative URL pattern routing
HttpServlet, HttpServletRequest, HttpServletResponse: Fundamental Java EE HTTP pipeline
doGet(request, response), doPost(request, response): Standard HTTP verb lifecycle methods
request.getSession(true), session.setAttribute(), session.getAttribute(): Session state persistence
session.invalidate(): Complete destruction of authenticated user sessions
request.getRequestDispatcher().forward(request, response): Server-side MVC view forwarding
response.setContentType(), response.setContentLengthLong(): HTTP binary header configuration
response.getOutputStream(), BufferedOutputStream: High-throughput binary response streaming
JSP Directives (<%@ page ... %>): Page imports, content type declarations, and error handling
JSP Scriptlets (<% ... %>) & Expressions (<%= ... %>): Dynamic server-side page composition

JDBC CONSTRUCTS USED:

Class.forName('com.mysql.cj.jdbc.Driver'): MySQL Connector/J driver class registration
DriverManager.getConnection(url, user, pass): Acquisition of active database connections
PreparedStatement, Connection.prepareStatement(sql): Pre-compiled SQL execution prevents injection
ResultSet, PreparedStatement.executeQuery(): Tabular query result set navigation and extraction
ResultSet.getInt(), getString(), getDouble(), getDate(): Strongly typed column value retrieval
try-with-resources / finally blocks: Deterministic closure of JDBC Statement and Connection instances

web.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
	xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
	xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
			http://xmlns.jcp.org/xml/ns/javaee/web-app_4_0.xsd"
	version="4.0">
```

```
  <display-name>Apex Student Profiles &amp;amp; Asset Streaming Portal</display-name>

  <!-- Welcome File Configuration -->
  <welcome-file-list>
    <welcome-file>login.jsp</welcome-file>
  </welcome-file-list>

  <!-- 1. Authentication Servlet -->
  <servlet>
    <servlet-name>AdminAuthServlet</servlet-name>
    <servlet-class>com.apex.portal.AdminAuthServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>AdminAuthServlet</servlet-name>
    <url-pattern>/adminAuth</url-pattern>
  </servlet-mapping>

  <!-- 2. Administrative Logout Servlet -->
  <servlet>
    <servlet-name>PortalLogoutServlet</servlet-name>
    <servlet-class>com.apex.portal.PortalLogoutServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>PortalLogoutServlet</servlet-name>
    <url-pattern>/portalLogout</url-pattern>
  </servlet-mapping>

  <!-- 3. Student Directory Servlet -->
  <servlet>
    <servlet-name>StudentDirectoryServlet</servlet-name>
    <servlet-class>com.apex.portal.StudentDirectoryServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>StudentDirectoryServlet</servlet-name>
    <url-pattern>/studentDirectory</url-pattern>
  </servlet-mapping>

  <!-- 4. Binary Media & Asset Streaming Servlet -->
  <servlet>
    <servlet-name>MediaStreamServlet</servlet-name>
    <servlet-class>com.apex.portal.MediaStreamServlet</servlet-class>
  </servlet>
  <servlet-mapping>
    <servlet-name>MediaStreamServlet</servlet-name>
    <url-pattern>/mediaStream</url-pattern>
  </servlet-mapping>

  <!-- Session Timeout Configuration (30 Minutes) -->
  <session-config>
    <session-timeout>30</session-timeout>
    <cookie-config>
      <http-only>true</http-only>
      <secure>false</secure>
    </cookie-config>
  </session-config>

</web-app>
```

DBConfig.java:

```
package com.apex.portal;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

/**
 * Thread-Safe JDBC Connection Manager (DBConfig.java)
 * Student: DARSHAN SUNDAR K K | Roll No: 24CS039
 */
public class DBConfig {
    private static final String DB_DRIVER = "com.mysql.cj.jdbc.Driver";
    private static final String DB_URL    =
"jdbc:mysql://localhost:3306/apex_college_admin?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
    private static final String DB_USER   = "root";
    private static final String DB_PASS   = "";

    // Base storage root directory for student attachments
    public static final String STORAGE_BASE = "C:/apex_portal_storage/";

    static {
        try {
            Class.forName(DB_DRIVER);
        } catch (ClassNotFoundException e) {
            System.err.println("Fatal: MySQL JDBC Driver failed to load -> " + e.getMessage());
        }
    }

    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
    }
}
```

AdminAuthServlet.java:

```
package com.apex.portal;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * Administrative Authentication Controller (AdminAuthServlet.java)
 * Student: DARSHAN SUNDAR K K | Roll No: 24CS039
 */
public class AdminAuthServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String username = request.getParameter("adminUser");
        String password = request.getParameter("adminPass");

        // Verify Administrator Credentials
        if ("admin".equalsIgnoreCase(username) && "admin123".equals(password)) {
```

```
            HttpSession session = request.getSession(true);
            session.setAttribute("adminPrincipal", username);
            session.setAttribute("role", "SYSTEM_ADMINISTRATOR");
            session.setAttribute("loginTime", new java.util.Date());

            // Redirect to Student Directory upon successful authorization
            response.sendRedirect(request.getContextPath() + "/studentDirectory");
        } else {
            // Forward back to login with contextual error notification
            request.setAttribute("authError", "Invalid administrator credentials. Access denied.");
            request.getRequestDispatcher("login.jsp").forward(request, response);
        }
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.sendRedirect("login.jsp");
    }
}
```

PortalLogoutServlet.java:

```
package com.apex.portal;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * Administrative Session Termination Controller (PortalLogoutServlet.java)
 * Student: DARSHAN SUNDAR K K | Roll No: 24CS039
 */
public class PortalLogoutServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session != null) {
            session.removeAttribute("adminPrincipal");
            session.removeAttribute("role");
            session.invalidate(); // Destroy session tokens
        }

        // Prevent browser caching of protected views
        response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
        response.setHeader("Pragma", "no-cache");
        response.setDateHeader("Expires", 0);

        request.setAttribute("logoutNotice", "Administrator session successfully terminated. Please sign in again.");
        request.getRequestDispatcher("login.jsp").forward(request, response);
    }
}
```

StudentDirectoryServlet.java:

```
package com.apex.portal;

import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

/**
 * Student Directory & Profile Forwarding Servlet (StudentDirectoryServlet.java)
 * Student: DARSHAN SUNDAR K K | Roll No: 24CS039
 */
public class StudentDirectoryServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        if (session == null || session.getAttribute("adminPrincipal") == null) {
            response.sendRedirect("login.jsp?error=unauthorized");
            return;
        }

        List<Map<String, Object>> studentList = new ArrayList<>();
        String sql = "SELECT s.student_id, s.roll_no, s.first_name, s.last_name, s.email, s.phone, " +
                     "s.gender, s.dob, s.cgpa, s.photo_path, s.doc_path, s.doc_original_name, " +
                     "s.enrollment_status, d.dept_code, d.dept_name " +
                     "FROM students s " +
                     "JOIN departments d ON s.dept_id = d.dept_id " +
                     "ORDER BY s.roll_no ASC";

        try (Connection conn = DBConfig.getConnection();
             PreparedStatement ps = conn.prepareStatement(sql);
             ResultSet rs = ps.executeQuery()) {

            while (rs.next()) {
                Map<String, Object> map = new HashMap<>();
                map.put("student_id", rs.getInt("student_id"));
                map.put("roll_no", rs.getString("roll_no"));
                map.put("fullName", rs.getString("first_name") + " " + rs.getString("last_name"));
                map.put("email", rs.getString("email"));
                map.put("phone", rs.getString("phone"));
                map.put("gender", rs.getString("gender"));
                map.put("dob", rs.getDate("dob"));
                map.put("cgpa", rs.getDouble("cgpa"));
                map.put("photo_path", rs.getString("photo_path"));
                map.put("doc_path", rs.getString("doc_path"));
                map.put("doc_orig", rs.getString("doc_original_name"));
                map.put("status", rs.getString("enrollment_status"));
```

```
            map.put("dept_code", rs.getString("dept_code"));
            map.put("dept_name", rs.getString("dept_name"));
            studentList.add(map);
        }
    } catch (Exception e) {
        request.setAttribute("dbError", "Failed to retrieve student directory: " + e.getMessage());
    }

    request.setAttribute("students", studentList);
    request.getRequestDispatcher("students.jsp").forward(request, response);
  }
}
```

MediaStreamServlet.java:

```
package com.apex.portal;

import java.io.BufferedInputStream;
import java.io.BufferedOutputStream;
import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Binary Media & Document Streaming Engine (MediaStreamServlet.java)
 * Streams images directly to <img> tags and serves PDF attachments dynamically.
 * Student: DARSHAN SUNDAR K K | Roll No: 24CS039
 */
public class MediaStreamServlet extends HttpServlet {
  private static final long serialVersionUID = 1L;

  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    String idParam   = request.getParameter("id");
    String assetType = request.getParameter("type"); // "photo" or "doc"

    if (idParam == null || assetType == null) {
      response.sendError(HttpServletResponse.SC_BAD_REQUEST, "Missing asset parameters.");
      return;
    }

    int studentId;
    try {
      studentId = Integer.parseInt(idParam);
    } catch (NumberFormatException e) {
      response.sendError(HttpServletResponse.SC_BAD_REQUEST, "Invalid identifier format.");
      return;
    }

    String filePath = null;
    String originalName = null;
```

```
String sql = "SELECT photo_path, doc_path, doc_original_name FROM students WHERE student_id = ?";
try (Connection conn = DBConfig.getConnection();
     PreparedStatement ps = conn.prepareStatement(sql)) {
   ps.setInt(1, studentId);
   try (ResultSet rs = ps.executeQuery()) {
      if (rs.next()) {
         if ("photo".equalsIgnoreCase(assetType)) {
            filePath = rs.getString("photo_path");
         } else if ("doc".equalsIgnoreCase(assetType)) {
            filePath = rs.getString("doc_path");
            originalName = rs.getString("doc_original_name");
         }
      }
   }
} catch (Exception e) {
   response.sendError(HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "Database lookup failed.");
   return;
}

if (filePath == null || filePath.trim().isEmpty()) {
   response.sendError(HttpServletResponse.SC_NOT_FOUND, "Asset not registered for this record.");
   return;
}

File file = new File(filePath);
if (!file.exists() || !file.isFile()) {
   response.sendError(HttpServletResponse.SC_NOT_FOUND, "Physical file unavailable on storage server.");
   return;
}

// Determine MIME Content-Type
String mimeType = getServletContext().getMimeType(file.getName());
if (mimeType == null) {
   String lower = file.getName().toLowerCase();
   if (lower.endsWith(".jpg") || lower.endsWith(".jpeg")) mimeType = "image/jpeg";
   else if (lower.endsWith(".png")) mimeType = "image/png";
   else if (lower.endsWith(".pdf")) mimeType = "application/pdf";
   else mimeType = "application/octet-stream";
}

response.setContentType(mimeType);
response.setContentLengthLong(file.length());

if ("doc".equalsIgnoreCase(assetType)) {
   String downloadName = (originalName != null && !originalName.isEmpty()) ? originalName : file.getName();
   response.setHeader("Content-Disposition", "inline; filename="" + downloadName + """);
} else {
   response.setHeader("Cache-Control", "public, max-age=86400"); // 1-day cache for images
}

// Stream binary contents using buffered I/O
try (BufferedInputStream bis = new BufferedInputStream(new FileInputStream(file));
     BufferedOutputStream bos = new BufferedOutputStream(response.getOutputStream())) {
   byte[] buffer = new byte[8192];
   int bytesRead;
   while ((bytesRead = bis.read(buffer)) != -1) {
      bos.write(buffer, 0, bytesRead);
   }
```

```
            bos.flush();
        }
    }
}
```

login.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Apex Administrative Portal - Sign In</title>
  <link rel="stylesheet" href="style.css">
</head>
<body class="login-body">
  <div class="login-container">
    <div class="login-card">
      <div class="login-header">
        <div class="institution-logo">&#x1F393;</div>
        <h2>Apex Institute of Technology</h2>
        <p class="login-subtitle">Centralized Student Records Management Portal</p>
      </div>

      <%-- Contextual Error Notifications --%>
      <% if (request.getAttribute("authError") != null) { %>
        <div class="alert alert-danger">
          &#x26A0; <%= request.getAttribute("authError") %>
        </div>
      <% } %>

      <% if (request.getAttribute("logoutNotice") != null) { %>
        <div class="alert alert-success">
          &#x2705; <%= request.getAttribute("logoutNotice") %>
        </div>
      <% } %>

      <form action="adminAuth" method="POST" class="login-form">
        <div class="form-group">
          <label for="adminUser">Administrator Username</label>
          <input type="text" id="adminUser" name="adminUser" placeholder="Enter username (admin)" required
autocomplete="username">
        </div>

        <div class="form-group">
          <label for="adminPass">Administrative Password</label>
          <input type="password" id="adminPass" name="adminPass" placeholder="Enter password (admin123)" required
autocomplete="current-password">
        </div>

        <button type="submit" class="btn btn-primary btn-block">&#x1F512; Authenticate &amp;amp; Access Portal</button>
      </form>

      <div class="login-footer">
        <p>&copy; 2026 Apex Institute of Engineering &amp;amp; Technology</p>
        <p class="text-muted">Protected System &bull; Unauthorized Access Prohibited</p>
      </div>
    </div>
  </div>
</body>
```

```
</html>
```

students.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.util.List" %>
<%@ page import="java.util.Map" %>
<%
   // Verify active administrative session
   if (session == null || session.getAttribute("adminPrincipal") == null) {
      response.sendRedirect("login.jsp?error=unauthorized");
      return;
   }
   List<Map<String, Object>> students = (List<Map<String, Object>>) request.getAttribute("students");
%>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Student Directory - Dynamic Asset Streaming</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="dashboard-wrapper">
    <header class="top-bar">
      <div>
        <h1>Apex Institute of Technology &bull; Student Dossiers</h1>
        <p class="tagline">Dynamic JDBC Data Retrieval &amp;amp; Servlet Image Streaming Engine</p>
      </div>
      <div class="user-control">
        <span class="user-badge">&#x1F464; <?= session.getAttribute("adminPrincipal") ?> (Admin)</span>
        <a href="portalLogout" class="btn btn-danger-sm">&#x23FB; Sign Out</a>
      </div>
    </header>

    <div class="portal-info-bar">
      <span>Active Database Scholars: <strong><%= (students != null) ? students.size() : 0 %></strong></span>
      <span>Asset Pipeline: <code>MediaStreamServlet</code> byte streaming via <code>&lt;img&gt;</code>
endpoints</span>
    </div>

    <!-- Student Cards Grid -->
    <div class="student-card-grid">
      <% if (students != null && !students.isEmpty()) {
          for (Map<String, Object> s : students) {
            Integer sId = (Integer) s.get("student_id");
            String rollNo = (String) s.get("roll_no");
            String name = (String) s.get("fullName");
            String deptCode = (String) s.get("dept_code");
            String deptName = (String) s.get("dept_name");
            Double cgpa = (Double) s.get("cgpa");
            String email = (String) s.get("email");
            String phone = (String) s.get("phone");
            String status = (String) s.get("status");
            String docPath = (String) s.get("doc_path");
      %>
        <div class="student-profile-card">
          <div class="card-avatar-wrap">
            <!-- Directly stream binary image from MediaStreamServlet -->
            <img src="mediaStream?id=<%= sId %>&type=photo" class="profile-avatar" alt="Avatar"
```

```
            onerror="this.onerror=null; this.src='https://ui-avatars.com/api/?name=<%= name
%>&background=0c2340&color=fff;">
          <span class="card-status status-<%= status.toLowerCase() %>"><%= status %></span>
        </div>

        <div class="card-details">
          <span class="card-roll"><%= rollNo %></span>
          <h3 class="card-name"><%= name %></h3>
          <p class="card-dept"><%= deptName %> (<%= deptCode %>)</p>

          <div class="card-metrics">
            <span class="metric-label">Cumulative GPA:</span>
            <span class="metric-val"><%= String.format("%.2f", cgpa) %> / 10.00</span>
          </div>

          <div class="card-contact">
            <div>&#x2709; <%= email %></div>
            <div>&#x1F4DE; +91 <%= phone %></div>
          </div>

          <div class="card-actions">
            <% if (docPath != null && !docPath.trim().isEmpty()) { %>
              <a href="mediaStream?id=<%= sId %>&type=doc" target="_blank" class="btn btn-stream-doc">&#x1F4C4;
Stream Verified ID PDF</a>
            <% } else { %>
              <span class="text-muted btn-disabled">&#x2718; No Document Attached</span>
            <% } %>
          </div>
        </div>
      </div>
    <%   }
      } else { %>
      <p class="empty-state">No student profiles loaded from the database.</p>
    <% } %>
  </div>
 </div>
</body>
</html>
```

style.css:

```
/* Java Servlets, JSP & Streaming Portal Stylesheet */
:root {
  --navy-dark: #0c2340;
  --navy-light: #1e3a8a;
  --amber-accent: #d97706;
  --bg-color: #f1f5f9;
  --card-bg: #ffffff;
  --border-color: #cbd5e1;
  --text-primary: #0f172a;
  --text-muted: #64748b;
  --font-family: 'Times New Roman', Times, serif;
}

* { box-sizing: border-box; margin: 0; padding: 0; }

body { font-family: var(--font-family); background-color: var(--bg-color); color: var(--text-primary); line-height: 1.45;
padding: 1.5rem; }
.login-body { display: flex; justify-content: center; align-items: center; min-height: 90vh; }
.login-container { width: 100%; max-width: 440px; }
```

```
.login-card { background: var(--card-bg); padding: 2.25rem; border-radius: 8px; border: 1px solid var(--border-color); box-shadow: 0 4px 15px rgba(12, 35, 64, 0.15); }

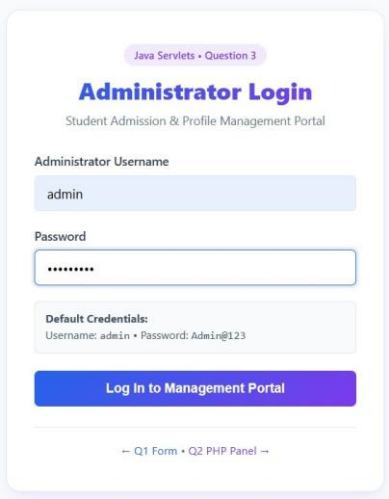
.login-header { text-align: center; margin-bottom: 1.5rem; }
.institution-logo { font-size: 2.5rem; margin-bottom: 0.5rem; }
.login-header h2 { font-size: 1.25rem; color: var(--navy-dark); }
.login-subtitle { font-size: 0.85rem; color: var(--text-muted); }
.login-form .form-group { margin-bottom: 1.25rem; display: flex; flex-direction: column; }
.login-form label { font-size: 0.88rem; font-weight: bold; margin-bottom: 0.35rem; }
.login-form input {
  font-family: var(--font-family);
  font-size: 0.92rem;
  padding: 0.6rem 0.8rem;
  border: 1.5px solid var(--border-color);
  border-radius: 5px;
}
.login-form input:focus {
  outline: none;
  border-color: var(--navy-light);
  box-shadow: 0 0 0 3px rgba(30, 58, 138, 0.15);
}
.login-footer { text-align: center; margin-top: 1.75rem; font-size: 0.8rem; color: var(--text-muted); }
.dashboard-wrapper { max-width: 1200px; margin: 0 auto; }
.top-bar {
  background: linear-gradient(135deg, var(--navy-dark) 0%, var(--navy-light) 100%);
  color: #ffffff;
  padding: 1.25rem 1.75rem;
  border-radius: 8px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1.25rem;
  border-bottom: 3px solid var(--amber-accent);
}
.top-bar h1 { font-size: 1.35rem; }
.tagline { font-size: 0.85rem; color: #cbd5e1; }
.user-control { display: flex; align-items: center; gap: 1rem; }
.user-badge { font-size: 0.9rem; background: rgba(255, 255, 255, 0.15); padding: 0.35rem 0.75rem; border-radius: 4px; }
.portal-info-bar {
  background-color: var(--card-bg);
  padding: 0.75rem 1.25rem;
  border-radius: 6px;
  border: 1px solid var(--border-color);
  margin-bottom: 1.5rem;
  display: flex;
  justify-content: space-between;
  font-size: 0.88rem;
}
.student-card-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(320px, 1fr));
  gap: 1.5rem;
}
.student-profile-card {
  background: var(--card-bg);
  border-radius: 8px;
  border: 1px solid var(--border-color);
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.06);
  padding: 1.25rem;
```

```
  display: flex;
  flex-direction: column;
}
.card-avatar-wrap { display: flex; justify-content: space-between; align-items: flex-start; margin-bottom: 1rem; }
.profile-avatar { width: 72px; height: 72px; border-radius: 8px; object-fit: cover; border: 2px solid var(--border-color); }
.card-status { padding: 0.2rem 0.55rem; border-radius: 4px; font-size: 0.75rem; font-weight: bold; }
.status-active { background: #ecfdf5; color: #047857; }
.status-graduated { background: #f0f9ff; color: #0284c7; }
.status-suspended { background: #fef2f2; color: #b91c1c; }
.card-roll { font-size: 0.85rem; font-weight: bold; color: var(--amber-accent); }
.card-name { font-size: 1.15rem; color: var(--navy-dark); margin: 0.2rem 0; }
.card-dept { font-size: 0.85rem; color: var(--text-muted); margin-bottom: 0.75rem; }
.card-metrics {
  background: #f8fafc;
  padding: 0.5rem 0.75rem;
  border-radius: 4px;
  margin-bottom: 0.75rem;
  display: flex;
  justify-content: space-between;
  font-size: 0.88rem;
}
.card-contact { font-size: 0.82rem; color: var(--text-muted); margin-bottom: 1rem; line-height: 1.4; }
.card-actions { margin-top: auto; border-top: 1px solid var(--border-color); padding-top: 0.85rem; }
.btn {
  font-family: var(--font-family);
  font-size: 0.88rem;
  font-weight: bold;
  padding: 0.5rem 1rem;
  border-radius: 5px;
  cursor: pointer;
  border: none;
  text-decoration: none;
  display: inline-block;
  text-align: center;
}
.btn-primary { background: var(--navy-dark); color: #ffffff; }
.btn-block { width: 100%; }
.btn-danger-sm { background: #dc2626; color: #ffffff; padding: 0.35rem 0.75rem; font-size: 0.8rem; border-radius: 4px; }
.btn-stream-doc { background: #e0f2fe; color: #0369a1; width: 100%; border: 1px solid #bae6fd; }
.btn-disabled { font-size: 0.82rem; display: block; text-align: center; }
.alert { padding: 0.75rem 1rem; border-radius: 6px; margin-bottom: 1.25rem; font-size: 0.88rem; }
.alert-danger { background: #fef2f2; color: #991b1b; border: 1px solid #fecaca; }
.alert-success { background: #ecfdf5; color: #065f46; border: 1px solid #a7f3d0; }
```

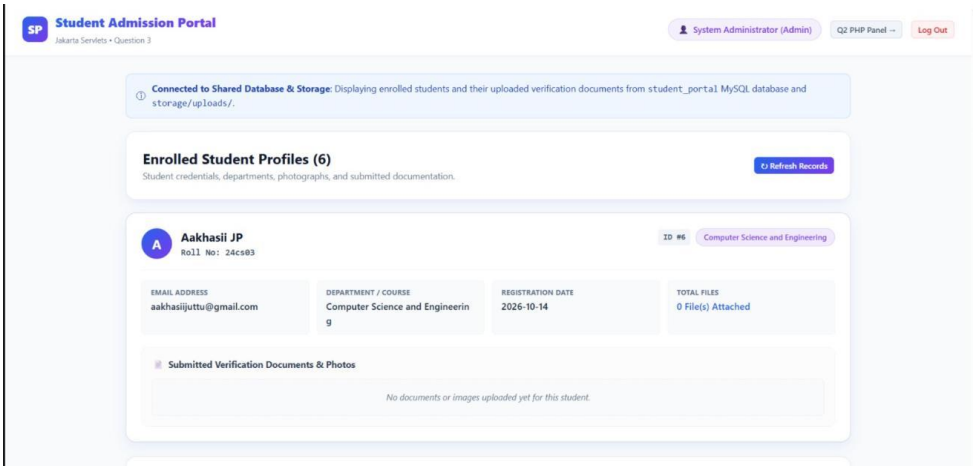
OUTPUT :

VALID OPERATIONS :

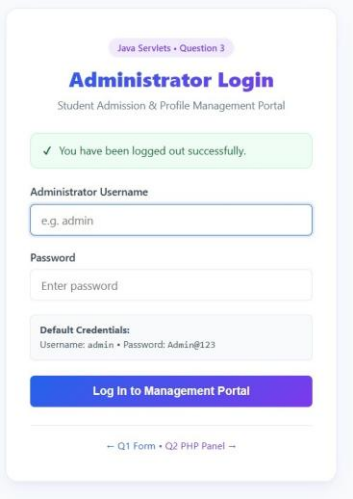
Valid Operation – Administrator Login Authentication :



Valid Operation – Verified Student Profile Dossiers with Servlet Streamed Assets :

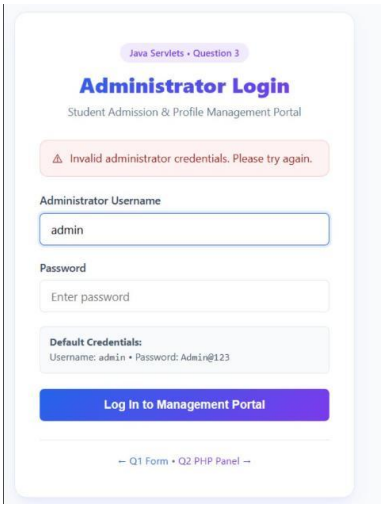


Valid Operation – Administrative Session Logout :



INVALID OPERATIONS :

Invalid Operation – Authentication Failure with Incorrect Credentials :



RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Servlet Architecture & web.xml Mapping (R1)	4	
JDBC Data Retrieval & Resource Management (R2)	4	
Servlet Binary Stream Handling (R3)	4	
Dynamic JSP Rendering & Conditional Logic (R4)	4	
Session Security & Authentication (R5)	4	
Total Marks	**20**	

RESULT:

A dynamic Java Web Portal was successfully developed and tested using Java Servlets, JDBC, and JSP. The system connects to MySQL via JDBC, streams binary photo assets directly to <img> tags using AssetStreamServlet, dynamically iterates over student profile records with JSP scriptlets, manages file attachment links conditionally, and enforces robust session security for administrative access.